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Forthcoming in
(2023) 109 Iowa L Rev

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LSE Law, Society and Economy Working Papers 6/2023
London School of Economics and Political Science
Law School

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Synthetic Data: Legal Implications of the Data-Generation Revolution

Michal S Gal* and Orla Lynskey**

Abstract: A data-generation revolution is underway. To date, most data used for algorithmic decision-making is collected from events that take place in the physical world (“collected” data) however this is about to change. By 2024, it is predicted that 60% of data used to train artificial intelligence systems globally will be synthetic. Synthetic data is artificially-generated data, created using generative AI, that has analytical value. It can be used to replace collected data by preserving or mimicking its properties or to supplement collected data to improve its completeness or to enhance privacy protections.

This article analyses the legal implications of synthetic data for markets and for society. It argues that synthetic data challenges the equilibrium found in existing laws which strike a balance between competing values, including data utility, privacy, security, and human rights. This article seeks to bring state-of-the-art data generation methods into the legal debate, and to propose legal reforms which capture the unique characteristics of synthetic data. While some of the challenges discussed here also arise with the use of collected data, synthetic data exacerbates these challenges putting further pressure on existing regulatory regimes and expediting the need for data governance reform.

Keywords: IT law, artificial intelligence law, regulation, big data, data privacy, personal data

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I. INTRODUCTION

Data is an essential input in our digital economies and societies.¹ Generally, the better the data (in terms of volume, variety, veracity, and velocity), the better the learning from it (information and knowledge). While algorithms and infrastructure are important elements of artificial intelligence (AI), data is a critical element in such value creation.²

Data is traditionally collected³ from the physical world (hereinafter: “collected data”).⁴ Some types of collected data are relatively abundant and easy to access and use, such as weather conditions. Yet many types of data are characterized by high access barriers.⁵ To be useful, data must be collected, cleaned and prepared, analyzed, and stored – any of which may be prohibitively costly or even impossible, at least for some.⁶ Accordingly, those who possess such data may enjoy comparative advantages, potentially providing them with data-based market power which can be exercised or even abused.⁷

But what if some types of data could be created without having to actively collect (all of) it from the real world? To overcome data access hurdles, data scientists make use of synthetic data: artificial data, generally generated by computer simulations or algorithms, which has analytical value.⁸ Such techniques use autonomous generation models or inferences from collected data. Synthetic data has numerous uses. First, it often mimics collected data, augmenting or replacing it. Such data is often used for training or testing machine learning AI;⁹ prominent examples include the voice recognition algorithm used in Amazon’s Alexa, and Google’s (now Waymo’s)

¹ JACQUES CRÉMER, YVES-ALEXANDRE DE MONTJOYE AND HEIKE SCHWEITZER, COMPETITION POLICY FOR THE DIGITAL ERA REPORT 73 (2019).

² See, e.g., Jeremy Khan, *Deep Learning Pioneer Andrew Ng Says Companies Should Get ‘Data-Centric’ to Achieve A.I. Success*, FORTUNE (June 21 2022), <https://fortune.com/2022/06/21/andrew-ng-data-centric-ai/>. The centrality of data to automated decision-making is also recognized in legal and ethical literature. See, e.g., Solon Barocas & Andrew D. Selbst, *Big Data’s Disparate Impact*, 104 CAL. L. REV. 671, 673-4 (2016); David Lehr & Paul Ohm, *Playing with the Data: What Legal Scholars Should Learn About Machine Learning*, 51 U.C. DAVIS L. REV. 653, 664 (2017); Brent Daniel Mittelstadt et al., *The Ethics of Algorithms: Mapping the Debate*, 3(2) BIG DATA & SOCIETY 1 (2016).

³ The selection of data to be collected and the characteristics of the collection method all affect the collected data, so it constitutes an incomplete representation of the real world. See generally “RAW DATA” IS AN OXIMORON (Lisa Gitelman ed., 2013); Marion Fourcade & Kieran Healy, *Seeing Like a Market*, 15 SOCIO-ECONOMIC REV. 9 (2017). The definition of collected data used in this article does not clash with this truism.

⁴ Different fields use different terms to define such data. These may include, for example, natural or original data.

⁵ See generally Daniel L. Rubinfeld & Michal S. Gal, *Access Barriers to Big Data*, 59 ARIZONA L. REV. 339, (2017).

⁶ *Id.*

⁷ *Id.*

⁸ We adopt the definition used by computer scientists. For the importance of maintaining analytical value see Donald B. Rubin, *Statistical Disclosure Limitation*, 9 J. OFF. STAT. 461, 462 (1993). Synthetic data is not to be confused with AI. While AI can be used to create synthetic data and can be trained on it, the two are not identical.

⁹ Elise Devaux, *Types of Synthetic Data and 5 Examples of Real-life Applications*, STATICE (Jan. 15, 2021), <https://www.statice.ai/post/types-synthetic-data-examples-real-life-examples>.

autonomous cars.¹⁰ Second, it is used to potentially reduce bias or to overcome statistical imbalance in representative examples; it may thereby increase the quality of decision-making.¹¹ Third, it is used to increase privacy or cybersecurity, thereby enabling wider use of valuable data for research and decision-making. For example, the U.S. Census Bureau transforms some of its data into synthetic data, in order to enable access.¹² Similarly, synthetic data can increase levels of privacy protection while enabling the operation of smart cities.¹³

While some forms of synthetic data generation have been around for some time (such as upsampling),¹⁴ the wide use of synthetic data in simulations to train machine learning models is quite new. This is enabled by recent advances in computing methods, computing power, and storage capabilities. As one observer noted, despite the fact that synthetic data generation technology is a relatively recent addition to the toolkit of machine learning engineers, “it has already evolved from the initially supportive role of augmenting real-world data, to one enabling a new wave of AI innovation”.¹⁵ This article focuses mainly on such uses.

Synthetic data has ingrained benefits. It can potentially reduce the costs involved in all stages of the data value chain, obviating the need for excessive data collection, costly cleaning and preparation, and long-term data storage. Generation-for-purpose may also shorten the time necessary to generate useful data. For these reasons, it is forecast that by 2024, 60% (!) of the data used to train AI systems around the world will be synthetic.¹⁶ Gartner predicted that by 2030, “you won’t be able to build high-quality, high-value AI models without synthetic data”.¹⁷ In many ways, synthetic data has the potential to do to data what synthetic threads did to cotton.

The importance of this data-generation revolution cannot be overstated. As we show, synthetic data may change the current balance between data utility and competing considerations. It also affects existing power relationships, both between competitors and between providers/suppliers and users/consumers. Synthetic data is thus poised to affect all spheres of our lives that involve data-based decision-making, including the economic, the social, and the political. Such effects on the welfare of individuals and

¹⁰ *Id.* Other examples involve facial recognition, detection of fraud and money laundering, and prediction of housing markets. *Id.*

¹¹ See *infra* Section II.

¹² *What are Synthetic Data?* U.S. CENSUS BUREAU (May 27, 2021), <https://www.census.gov/library/fact-sheets/2021/what-are-synthetic-data.html>.

¹³ Alex LaCasse, *Synthetic data a Key to Privacy by Design Practices in New Canadian Smart City Partnership*, IAPP (Nov. 29, 2022), <https://iapp.org/news/a/synthetic-data-is-key-to-privacy-by-design-practices-in-new-canadian-smart-city-partnership/>.

¹⁴ The process of adding more data points of a certain type to the dataset, usually to reduce unbalanced sampling. Nandhini Nallamuthu, *Handling Imbalanced Data – Machine Learning, Computer Vision and NLP*, ANALYTICS VIDHYA (Nov. 7, 2020), <https://www.analyticsvidhya.com/blog/2020/11/handling-imbalanced-data-machine-learning-computer-vision-and-nlp/>.

¹⁵ Andrey Shtylenko, *The Advantages of Synthetic Data, The Reality Gap*, LINKEDIN (Nov. 23, 2022), https://www.linkedin.com/pulse/advantages-synthetic-data-andrey-shtylenko?trk=news-guest_share-article.

¹⁶ Andrew White, *By 2024, 60% of the Data Used for the Development of AI and Analytics Projects will be Synthetically Generated*, GARTNER (July 24, 2021), https://blogs.gartner.com/andrew_white/2021/07/24/by-2024-60-of-the-data-used-for-the-development-of-ai-and-analytics-projects-will-be-synthetically-generated/.

¹⁷ Leinar Ramos & Jitendra Subramanyam, *Maverick* Research: Forget About Your Collected data – Synthetic Data Is the Future of AI*, GARTNER (June 24, 2021), <https://www.gartner.com/en/documents/4002912>.

societies may be both positive and negative. While many of these effects also arise with regard to collected data, synthetic data puts them on steroids.

Given the importance of this revolution, this article seeks to identify and critically examine the effects of synthetic data on key data governance challenges. In particular, it focuses on three main issues that stand at the basis of our legal data regime, and that are significantly affected by synthetic data: **data access**, **data privacy**, and **data accuracy**. For each, it queries whether the existing legal regime is fit for purpose, able to address the governance challenges through either current application or a modified interpretation of the law, or whether the regulatory toolbox should be updated.

With respect to the first issue, by lowering some access barriers to data, synthetic data affects data access and data flows, potentially changing the competitive dynamics in markets where such access constitutes a significant barrier. Importantly, synthetic data can potentially overcome comparative advantages resulting from data-based network effects and feedback loops, which allow entities that already possess large datasets to keep ahead on the data learning curve, continually improving their decision-making relative to others¹⁸ and entrenching their data-based market power.¹⁹ In some circumstances synthetic data can potentially break such self-perpetuating loops. There is, of course, the countervailing risk that, in cases where collected data is unique and essential for synthetic data generation, the advent of synthetic data could instead increase the comparative advantages of those controlling collected data. However, such instances are becoming less common.²⁰

Despite its potential, the effects of synthetic data on competition have not, as of yet, been recognized by academics, legislators, and regulators. Take, for example, calls for stringent regulation of large digital firms. As elaborated below, new regulations are being suggested, and cases are being brought, based on assumptions of data-based market power that are no longer true for some markets.²¹ Some proposed regulations that may already be obsolete in certain markets include mandatory data sharing, portability, interoperability, and standardization.²² Recognizing the effects of synthetic data may lead to a more nuanced hands-off regulatory approach in this legal realm.

In contrast, the effects of synthetic data on meeting the requirements of data privacy laws have been acknowledged. Yet most of the discussion around this issue so far has been misleading or simplistic. For instance, it is often argued that synthetic data – based on transformations of collected data – can overcome the constraints on data use imposed by privacy laws.²³ Implicit in these claims is the assumption that synthetic data processing does not pose a risk to privacy and related rights, such as autonomy,

¹⁸ UK DIGIT. COMPETITION EXPERT PANEL, UNLOCKING DIGITAL COMPETITION 33 (2019), https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/785547/unlocking_digital_competition_furman_review_web.pdf.

¹⁹ STIGLER CTR. FOR THE STUDY OF THE ECON. & THE STATE, STIGLER COMM. ON DIGIT. PLATFORMS: FINAL REPORT 35-6 (2019), <https://www.chicagobooth.edu/-/media/research/stigler/pdfs/digital-platforms---committee-report---stigler-center.pdf>

²⁰ SERGEY I. NIKOLENKO, SYNTHETIC DATA FOR DEEP LEARNING 8 (2021).

²¹ See *infra*, Section III.B.

²² *Id.*

²³ *Anonymization and Data Privacy with Static*, STATICE.AI (last visited Jan. 23, 2023), https://privacy.static.ai/hubfs/Resources/brochures/Anonymization_data_privacy_Static.pdf.

non-discrimination, and human dignity. Yet, as we show, this is not the case. To the contrary: synthetic data may sometimes exacerbate such harms, both amplifying traditional harms and creating novel ones. One extreme example involves deep fakes that are presented as real, in which at least part of the data used to create fake images is synthetic.²⁴ But, more commonly and thus potentially more importantly, synthetic data increases data externalities and collective data harms. In particular, it accentuates the fact that not only is data itself intangible and non-rivalrous, but that the learnings from data also share such characteristics. Yet, as we show below, such effects are often not regulated by privacy laws.

Synthetic data can also potentially increase the quality of a dataset, thereby raising the quality of data-based decision-making. Improved data quality amplifies the benefits, but also the risks, associated with more granular data.²⁵ The latter include, inter alia, a better ability to profile, nudge, exploit and manipulate individuals, with ramifications for the interpersonal, commercial, social, and political spheres.²⁶ As we show, the legal framework that currently applies to increased data quality does not create an optimal balance between the competing considerations, especially once synthetic data is added to the mix. Most laws that relate to data quality mandate *increased* accuracy, rather than less; and many harmful uses of accurate data are not regulated.

This reality requires us to reevaluate our regulatory tools, which were designed with collected data in mind. To name but a few challenges, it calls upon us to consider a shift in the focus of data governance models from data collection to its uses and effects; from user consent and control to notions of welfare and well-being; and from private data to inference data and to collective data harms. It also requires us to rethink the current balance between data utility, privacy, security, and human rights, and the tools currently used to protect this balance. For example, it challenges the explainability requirements for AI-based decisions, potentially moving the focus from causality to generation-process reliability;²⁷ and it affects the interpretability of reasonableness principles in contracts and laws pertaining to the collection and use of data. This article takes a first step in this direction.

We begin in Section 2 by providing a brief overview of synthetic data: its definition, main production techniques, costs and benefits, and how it is being used in practice. We then focus on the three main effects of synthetic data: data access, data privacy, and data quality. Section 3 charts its effects on competition and market dynamics, which, in turn, affect the functioning of markets in the data economy. Section 4 analyzes its impact on data privacy. Section 5 focuses on the broader societal implications of more granular, complete, and representative synthetic data. All sections address the extent to which the existing legal landscape responds adequately to the changes wrought by synthetic data. As such, the article adds to the growing literature calling into question the ability of existing legal frameworks to respond to the challenges of new data-related

²⁴ Danielle K. Citron & Robert Chesney, *Deep Fakes: A Looming Challenge for Privacy, Democracy and National Security*, 107 CAL. L. REV. 1753, 1760 (2019).

²⁵ For an excellent discussion of algorithmic accuracy, which has implications for data accuracy, see Aileen Nielsen, *The Too Accurate Algorithm* (2022), <https://www.research-collection.ethz.ch/handle/20.500.11850/572429>

²⁶ See *infra* Section V.

²⁷ See *infra* Section V.

techniques. As we show, a new balance is needed, which would allow us to enjoy the significant competition, innovation, privacy, security, human rights, and data quality benefits that this data revolution creates, while addressing the concerns it raises in these spheres. Section 6 concludes and offers avenues for further research.

II. TECHNOLOGICAL BACKGROUND

Opening the synthetic data “black box” and looking under its hood is essential for determining whether our current laws can ensure that its uses increase social welfare. Accordingly, this section explores what synthetic data is, how it is generated, and its benefits and limitations relative to collected data.

A. WHAT IS SYNTHETIC DATA?

Synthetic data is artificially generated data with analytical value.²⁸ Its generation can be based on collected data, or on assumptions made by the coder (be it a human or AI) about the different variables in the dataset.²⁹ Accordingly, the main difference between collected and synthetic data is their source: while the former is collected from the real world, via human or technological sensors, the latter is generated artificially.

Synthetic data generation is a general-purpose technology that can be employed in numerous spheres and industries, including health, transportation, and finance.³⁰ Its flexibility is also reflected in the variety of outputs that it can give rise to, including datasets, images, audio files, and videos. The resulting datasets can be fully or partially synthetic. While synthetic data was first created in 1993, it has been significantly developed since then, partly due to advancements in technologies for processing, analyzing, and storing data.³¹

B. SYNTHETIC DATA GENERATION

Numerous methods for generating synthetic data exist. We group these methods into three categories, which are distinguished by their need for collected data (which could be public, private, or a combination thereof) in the generation process. This parameter enables us to explore the extent to which data collection barriers can be overcome, as well as the potential use of private data in the generation process, two conditions that affect our analysis below. The first group requires the same amount of

²⁸ See *supra* footnote 8.

²⁹ Rubin, *supra* note 8.

³⁰ Harvard Business Review Analytics Service, *Briefing Paper, The Executive's Guide to Accelerating AI and Data Innovation with Synthetic Data* (2021), <https://f.hubspotusercontent20.net/hubfs/4408323/HBR%20campaign/HBR%20Analytic%20Services%20Synthetic%20Data.pdf>.

³¹ See, e.g., *Accelerating AI with Synthetic Data*, NVIDIA, https://www.nvidia.com/content/dam/en-zz/Solutions/deep-learning/resources/accelerating-ai-with-synthetic-data-ebook/accelerating-ai-with-synthetic-data-nvidia_web.pdf (last visited Jan. 23, 2023).

collected data as traditional methods used to create similar outputs. The second reduces the amount or the quality of collected data necessary. The third does not require any direct use of collected data. While the different methods can be combined in the data generation pipeline, here we relate to them in their pure form, to emphasize their distinctive characteristics. In each example we also explore their main uses and whether new information – defined here as information that could not be directly learned from the input data used in its generation – is created. Our analysis mainly focuses on synthetic data that is generated by machine learning algorithms, and that can be created *en masse*.

1. Generation Based on Transformations of Collected Data

Synthetic data can be generated based on transformations of collected data. While such data serves important goals – most importantly enabling better **extraction** of information from the collected data, or wider **sharing** of the data by deidentification – it does not significantly reduce the need for collected data or change the features such data must contain. This group of methods have been used for quite some time.³²

Synthetic data is often generated as a stage in the data flow pipeline, geared towards extracting more information from collected data. Consider two examples. In the first, as collected data is cleaned and prepared for analysis, some of the values in the dataset are replaced with synthetic values in order to ensure consistency (a process called data curation).³³ For instance, if most data points relate to minutes, those that relate to hours can be replaced by synthetic data to correct the inconsistency. In the second example, the synthetic data results from the analysis performed on the collected data. Take, for example, *collaborative filtering*, which is based on computing similarities between clusters. To illustrate, assume that a seller wishes to create a dataset capturing the probability that a buyer might be interested in different items.³⁴ The coder first identifies two types of clusters in the collected data: clusters of relatively similar users; and clusters of relatively similar items. He then calculates two types of attributes: attributes of users (e.g., demographic information), and attributes of items (e.g., film genres). The algorithm then determines the distribution of user clusters and item clusters and affiliates them with user/item attributes respectively. These generated correlations create a partially synthetic dataset, where synthetic data fills in the gaps in the collected dataset, enabling learnings from the input data to be more easily extracted or conveyed.

More interesting for our purposes is the generation of new, wholly synthetic datasets, based on transformations of the collected data. Such transformations serve many useful purposes, including **transfer** or **storage** of data where costs depend on the volume of the data, or **deidentification** of data. The basic idea behind this method is quite simple: computing the (main) statistical characteristics of the original dataset, and creating a synthetic one, with quite similar characteristics. It involves the following main

³² See, e.g., Rubin, *supra* note 8, at 461; NIKOLENKO, *supra* note 20, at 139-160.

³³ Data Curation, TECHTARGET (last visited Jan. 23, 2023), <https://www.techtarget.com/searchbusinessanalytics/definition/datacuration#:~:text=Data%20curation%20is%20the%20process,group%20or%20the%20general%20public>.

³⁴ Karen Tso & Lars Schmidt-Thieme, *Attribute-Aware Collaborative Filtering*, in FROM DATA AND INFORMATION ANALYSIS TO KNOWLEDGE ENGINEERING: STUDIES IN CLASSIFICATION, DATA ANALYSIS, AND KNOWLEDGE ORGANIZATION 614 (M. Spiliopoulou et al. eds., 2006).

steps.³⁵ The first is data preparation: cleaning the collected data to remove errors, ensuring that all fields in the dataset use consistent coding schemes, and confirming that data from multiple sources is mapped into the same data typology.³⁶ The next step is developing a Data Generator to generate synthetic data based on manipulations of the collected data. The Generator's algorithm computes the metrics for the collected data, and then sets the parameters that will be used to generate synthetic data. To maintain logical consistency, some characteristics of the original dataset may need to be checked (for example, no man can have a positive value for "pregnant"). The third step is computing metrics for the synthetic data. Finally, the metrics of the collected and the synthetic data are compared using a Discriminator. This step assesses the utility of the synthetic dataset by determining whether its statistical properties are (relatively) similar to those of the original set.³⁷ If this comparison concludes that the synthetic data is too different from the collected data, the generation parameters are adjusted and new synthetic data is generated. The process iterates until acceptable synthetic data is produced.³⁸ These utility comparisons can be formalized using various similarity metrics that are repeatable and automated.³⁹ An additional, optional step involves a feedback loop which refines the Generator to improve the synthetic data against the relevant comparison metrics. Where privacy concerns arise, a privacy assurance assessment can be added, to ensure that privacy risks remain below a certain benchmark.

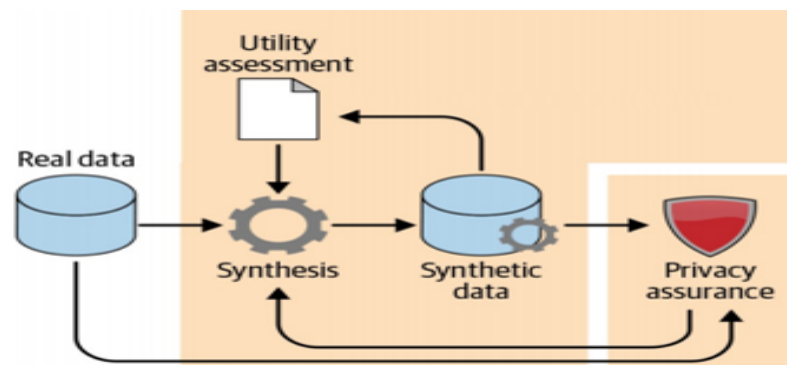


Figure 1: Data generation process when collected data is used as an input⁴⁰

The Data Generator can use different techniques, such as decision trees or deep learning.⁴¹ The exact choice is driven by the characteristics of the collected data, including its complexity, and the level of data utility desired.⁴² A common model

³⁵ *Synthetic Data*, JPMORGAN, <https://www.jpmorgan.com/technology/artificial-intelligence/initiatives/synthetic-data> (last visited Jan. 23, 2023).

³⁶ *Id.*

³⁷ NVIDIA, *supra* note 31, at 12.

³⁸ *Id.*, at 13-4.

³⁹ *Id.*

⁴⁰ KHALED EL EMAM, LUCY MOSQUERA & RICHARD HOPTROFF, PRACTICAL SYNTHETIC DATA GENERATION: BALANCING PRIVACY AND THE BROAD AVAILABILITY OF DATA (2020), figure 2.14.

⁴¹ NIKOLEKO, *supra* note 20, at Section 4.1.

⁴² *Neural Network*, INVESTOPEDIA, <https://www.investopedia.com/terms/n/neuralnetwork.asp> (last visited Jan. 23, 2023).

involves variational autoencoders (VAE).⁴³ VAE is an unsupervised machine learning method which results in a meaningful representation of a multidimensional dataset. VAEs function in two steps. First, an encoder network transforms an original complex distribution into a more compact representation with fewer dimensions. Second, the decoder then takes that compressed representation and reconstructs the original input data, generating synthetic data. The VAE is trained by optimizing the similarity between the synthetic data and the input data.⁴⁴ VAEs are relatively **easy to implement and to train**. Yet as the original data becomes more heterogeneous, it becomes more **difficult to formulate a reconstruction method that works well on all data points**.⁴⁵

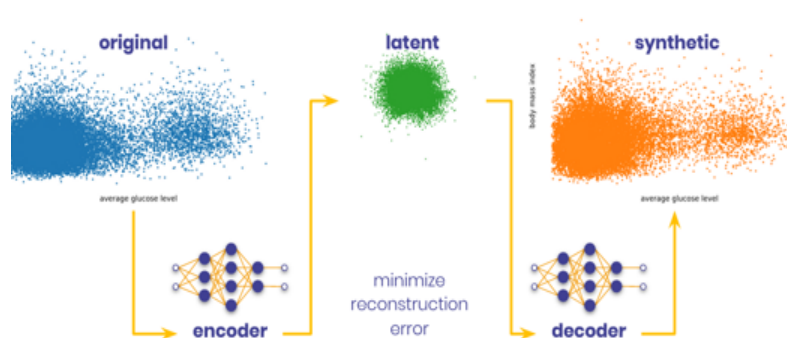


Figure 2: Variational autoencoders⁴⁶

Common to all such techniques is the fact that the information which can be learned from the synthetic dataset is wholly based on the input data.

2. Generation Methods Which Reduce the Need for Collected Data

The second group of generation methods are those which reduce the amount or quality, or change the features, of collected data necessary in order to achieve a given result. A prominent example of this group is generative adversarial networks (GAN), which employs two neural networks (deep learning algorithms), pitting them against each other in an adversarial fashion, in a zero-sum game. The first network, the Generator, generates synthetic data without directly using the collected data (this process is explained in the next section). The generated data is then sent to the second neural network – the Discriminator – which was trained on collected data. The Discriminator compares the synthetic data with the collected data, creating a propensity score and determining which parts of the data give away its “fakeness”. The result is then fed back to the Generator. A good synthetic model is created when the Discriminator is unable to distinguish between the collected and synthetic datasets. The main weakness of GANs is that they are more challenging to train than VAEs.⁴⁷

⁴³ *How to Generate Synthetic Data*, STATICE, <https://www.statice.ai/post/how-generate-synthetic-data> (last visited Jan. 23, 2023).

⁴⁴ EL EMAM ET AL., *supra* note 40, at 128.

⁴⁵ *How to Generate*, *supra* note 43.

⁴⁶ *Id.*

⁴⁷ *Id.*

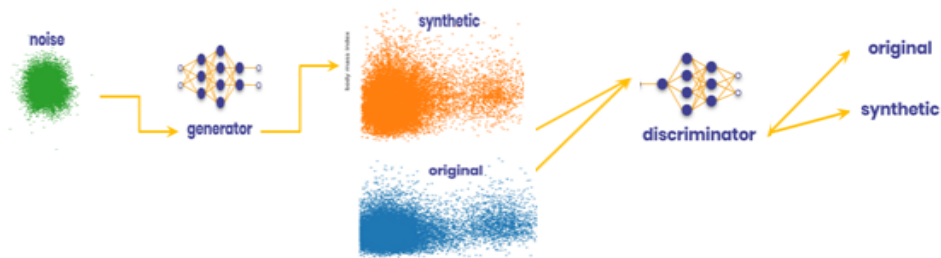


Figure 4: Generative Adversarial Networks⁴⁸

This approach is particularly useful for **synthetic image generation**.⁴⁹ For example, assume that a Generator has created a fake image of a stop sign. The Discriminator might determine that the image is fake based on mismatches between the fake image and collected data available to the algorithm (e.g., the color of the fake image may not be coherent with the algorithm’s learning from the collected data which conveys the actual color of stop signs).⁵⁰ Such feedback is fed into the Generator, which generates an updated image. The process repeats until the Discriminator can no longer identify that the image is fake.

Another use is **upsampling** – a technique for increasing the granularity or quality of an output by adding new data points between existing points.⁵¹ Upsampling can be used to reduce bias that might result from training an automated system on biased or incomplete datasets. To illustrate, consider Amazon’s famous attempt to train an algorithm to rate job applicants for technical posts. The algorithm was trained on resumes submitted to the company in previous years, which reflected male dominance in the industry. The result was that it judged male applicants as superior, and penalized references in resumes which indicated the applicant was a woman (e.g., women’s football captain).⁵² Upsampling could counter such bias by adding synthetic data representing the resumes of successful female applicants. Those can be created by a Generator and vetted by a Discriminator which learned from a few real resumes of successful women.

Both of these uses extend the available dataset with transformations that do not change the properties that one wishes to learn (“data augmentation”).⁵³ Accordingly, the information that can be directly learned from the synthetic data created is wholly based on the input data (e.g., the data does not suggest new characteristics of stop signs). At the same time, new information can be learned from the *process* of creating the synthetic data (such as what elements in the collected data are the most efficient differentiating factors of stop signs, or what elements can “trick” the Discriminator).

⁴⁸ *Id.* The term “original” can refer to “collected.”

⁴⁹ For example, the website <https://thispersondoesnotexist.com/> includes images of people who do not exist, created by GAN. It is impossible for a human observer to determine whether the image is real.

⁵⁰ EL EMAM ET AL., *supra* note 40.

⁵¹ See *supra* fn. 14.

⁵² Jeffrey Dastin, *Amazon Scraps Secret AI Recruiting Tool that Showed Bias Against Women*, REUTERS (Oct. 11, 2018), <https://www.reuters.com/article/us-amazon-com-jobs-automation-insight-idUSKCN1MK08G>

⁵³ NIKOLENKO, *supra* note 20, at 6.

This new learning can help reduce training and error costs. Such synthetic outcomes are often used as inputs for simulations, to which we turn next.

3. *Synthetic Data Generation without (Direct) Use of Collected Data*

Some types of synthetic data can be generated without (direct) use of collected data. This is done via a data simulator. Such a simulator generates synthetic data based on a set of rules which determine the relationships between the relevant data attributes.⁵⁴ The complexity of the generation method chosen is affected by the potential use of the dataset. As noted, such simulators have recently become the main tool for **training and testing machine learning algorithms**.⁵⁵

Some simulators do not require collected data at all. Consider a simple example: creating a dataset of numbers to train an algorithm to organize numbers sequentially. A synthetic data simulator which picks numbers randomly will suffice.

Despite the fact that they do not require collected data, such simulations can create new information. To illustrate, let us contrast two well-known examples. IBM's Deep Blue trained an algorithm to play chess by feeding it numerous examples of winning strategies (collected data).⁵⁶ The algorithm famously defeated world chess champion Garry Kasparov in a six-game match.⁵⁷ Contrast this with AlphaGo, developed by DeepMind – a subsidiary of Google (now Alphabet) – which was taught to play the board game Go.⁵⁸ The algorithm was first fed the rules of the game. It was mainly trained by creating numerous simulations of games where the algorithm played against other instances of itself, using reinforcement learning to improve its play.⁵⁹ Each algorithm was, in effect, creating the synthetic dataset of moves from which the other algorithm learned.⁶⁰ AlphaGo proved itself by defeating human world champions.⁶¹ But, more importantly, it developed new strategies for playing Go, thereby adding new learning.⁶²

Also interesting for our analysis are cases in which collected data is indirectly used, in that the simulation model relies on prior exposure of the coder (human or AI) to such data (i.e., background knowledge). The synthetic data is then based on the coder's assumptions regarding the statistical properties of the relevant data attributes.⁶³ For example, the coder may base the maximum speed humans are shown reaching in synthetic videos on his real-world observations of human locomotion. Background knowledge can also be based indirectly on collected data, such as when the learning

⁵⁴ NVIDIA, *supra* note 31, at 4, 8-9.

⁵⁵ Ramos & Subramanyam, *supra* note 17.

⁵⁶ *Deep Blue*, WIKIPEDIA (last visited Jan. 23, 2023), [https://en.wikipedia.org/wiki/Deep_Blue_\(chess_computer\)](https://en.wikipedia.org/wiki/Deep_Blue_(chess_computer)).

⁵⁷ *Id.*

⁵⁸ AlphaGo, WIKIPEDIA (last visited January 23, 2023), <https://en.wikipedia.org/wiki/AlphaGo>.

⁵⁹ While it was first given examples of moves by expert players from recorded historical games (collected data), this was mainly done to shorten the time the algorithm needed to reach a certain level of proficiency in the game. *Id.*

⁶⁰ David Silver & Demis Hassabis, *AlphaGo: Mastering the Ancient Game of Go with Machine Learning*, GOOGLE RESEARCH BLOG (Jan. 27, 2016), <https://ai.googleblog.com/2016/01/alphago-mastering-ancient-game-of-go.html>.

⁶¹ AlphaGo, *supra* note 58.

⁶² *Id.*

⁶³ EL EMAM ET AL., *supra* note 40.

from collected data in another context is embedded in the model (transfer learning). Take an example from retail: companies can use 3D simulations to create a synthetic dataset containing a thousand images from as few as five actual images of a product.⁶⁴ To do this, they employ existing knowledge about how different shapes look from different angles, learned from different tasks.⁶⁵

4. Typology of Synthetic Datasets

The analysis above serves as a basis for the following typology of synthetic data, based on the type of input data used and the resultant synthetic data output:

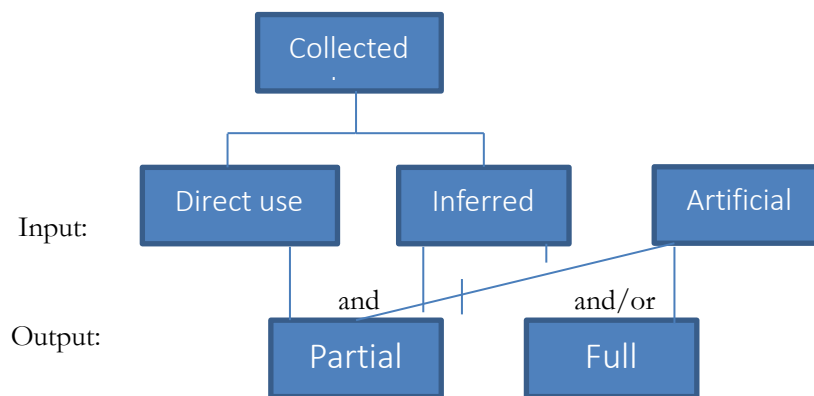


Figure 5: A Typology of Datasets

This typology also echoes the fact that different types of data may relate to different stages in the data processing pipeline, creating a causal chain whereby (temporary) datasets influence each other.⁶⁶ For example, collected data can be used as a (partial) basis for creating synthetic data, which can then be used in simulations that create a different form of synthetic data. A good example is training autonomous vehicles: a small set of collected images is used either in the Generator or the Discriminator to create a large number of synthetic images, which are relevant for simulations of road conditions for machine learning training purposes.

⁶⁴ Gerard Andrews, *What is synthetic data?*, NVIDIA (June 8, 2021), <https://blogs.nvidia.com/blog/2021/06/08/what-is-synthetic-data/>. These datasets are part of a range of technologies, including computer vision and geofencing, which underlie smart stores – physical stores where customers pay for goods via an app rather than interacting with a cashier or checkout machine.

⁶⁵ *Id.*

⁶⁶ Sebastian Benthall, *Situated Information Data Flows*, in *Hot Topics in the Science of Security Symposium* (ACM), April 1–3, 2019, <https://doi.org/10.1145/3314058.3314066>.

B. BENEFITS AND COSTS OF SYNTHETIC DATA

Why is artificial data generation used? We identify three main reasons that provide a wider context for the generation methods just explored.

The first reason, which is relevant to the second and third types of generation methods explored above, is that it allows for *replacing collected data characterized by high access barriers*. While synthetic data is not a panacea, it may reduce data-related barriers in all parts of the data value chain: collection–preparation–analysis–storage–use. By enabling generation-for-use, synthetic data can lower the costs and the time involved in collecting the relevant data. This is especially important given that machine learning algorithms are generally data-intensive,⁶⁷ and many types of datasets are expensive or hard to find.⁶⁸ Also, as Fromer notes, cloud computing, machine learning, and automation of tasks increase the secrecy of input data, thereby making it harder to access indirectly.⁶⁹ To overcome such barriers, firms might be able to produce synthetic data internally, or obtain it from third parties who specialize in such data production.⁷⁰

Synthetic data also reduces the resources needed for preparation of the raw data for analysis, which involves, inter alia, cleaning, labelling, and organizing the raw data. Such tasks can be complex, laborious, and expensive.⁷¹ In particular, manual labelling is often costly, time-consuming, and error-prone.⁷² By labelling and organizing the data automatically during the generation process,⁷³ synthetic data combines data collection and preparation, creating data that is fit for purpose from the start. This is especially important for machine learning algorithms, where the scale of datasets can reach hundreds of thousands and even millions of data points. One entrepreneur estimated that “a single image that could cost \$6 from a labeling service can be artificially generated for six cents”.⁷⁴ In the data analysis stage, synthetic data can be used to make the analysis more efficient. Interestingly, synthetic data can be used to train algorithms so that they make synthetic data more suitable for training.⁷⁵

Synthetic data can also potentially reduce storage costs, in four main ways. First, if a synthetic dataset can be easily recreated, its user does not need to store data for future use.⁷⁶ Indeed, synthetic data can be generated only when needed (what data scientists call “lazy production”). Second, and relatedly, generation-for-purpose reduces the need to store data for long periods before a sufficient amount is accumulated for meaningful analysis. Third, generation-for-purpose reduces the amount of redundant data that

⁶⁷ Open data sources are often limited in their availability or utility. Marco Iansiti, *The Value of Data and Its Impact on Competition* (Working Paper No 22-002, Harvard Business School, 2021).

⁶⁸ NIKOLENKO, *supra* note 20, at 12.

⁶⁹ Jeanne C. Fromer, *Machines As the New OompaLoompas: Trade Secrecy, the Cloud, Machine Learning, and Automation*, 94 NYU L. REV. 706, 723 (2019). See also Lehr & Ohm, *supra* note 2, at 655.

⁷⁰ Markets exist for several types of synthetic data. See, e.g., Elise Devaux, *List of Synthetic Data Startups and Companies — 2021*, MEDIUM (March 23, 2021), <https://elise-deux.medium.com/the-list-of-synthetic-data-companies-2021-5aa246265b42>.

⁷¹ While there exist large open datasets for many uses (such as ImageNet), many of them are only labelled for certain uses, or may contain inherent labelling biases. NIKOLENKO, *supra* note 20, at 2.

⁷² Shtylenko, *supra* note 15.

⁷³ *Id.*

⁷⁴ Andrews, *supra* note 64.

⁷⁵ NIKOLENKO, *supra* note 20, at 13.

⁷⁶ For such barriers see Rubinfeld & Gal, *supra* note 5, at 363-4.

might otherwise be included in the dataset. This is especially important where the relevant information cannot easily be parsed from collected datasets that take up a lot of memory, as in the case of videos. Finally, it can obviate the need for storage in cases where the user does not know a priori precisely what kinds of data might be needed for his analysis, yet such data can be synthetically generated in the future.

The generation of synthetic data might also reduce obstacles to use, in that some limitations – such as legal prohibitions on certain uses of personal data – may not apply to synthetic data, a point we return to later.⁷⁷ Furthermore, where it can be internally rather than externally generated, synthetic data can overcome some technical or legal obstacles to data transfers.⁷⁸

Importantly, to replace collected data, synthetic data need not be similar to it. Indeed, for some purposes optimal results might require using synthetic data which does not reflect actual real-world conditions.⁷⁹ For example, a dataset used to train autonomous vehicles may stimulate faster and more effective learning if it contains an outsized proportion of risky situations, such as people jumping into the road.

Second, synthetic data can *enrich the data pool with new or higher quality data*, which can augment or replace collected datasets. Such synthetic data allows analysts to study phenomena for which a sufficient amount of collected data cannot be (affordably) collected, or where collected data is not easily available for use (e.g., the collected data is not labelled, or is labelled correctly).⁸⁰ Importantly for our analysis, data augmentation may enable the extension of small, context-specific datasets in a way which does not alter essential underlying features of the data, yet creates new, relevant data (as in the case of 3D simulations mentioned above).⁸¹

On its face, generating such data seems like a cheap trick. If you can learn correlations from collected data, then why do you need to apply it to more data? Would this not simply generate similar results? Part of the answer is that even if we know all the theoretical parameters in a dataset, *value can be found in creating a new dataset which makes use of interactions between parameters in simulations*. A notable recent example involves the algorithm-generated solution to the protein-folding problem. In 2020, DeepMind unveiled AlphaFold, which uses computer simulations, based on background knowledge about proteins, to accurately and efficiently predict the 3D shape of any unknown protein using just its DNA or RNA source code.⁸² AlphaFold's predictions are so accurate that the protein-folding problem is considered solved after more than 70 years of searching; and its open-access database now contains over 200 million predicted protein structures. The tool and database together comprise a significant

⁷⁷ See discussion in Section IV *infra*. It might be interesting to explore how the possibility of generating synthetic data able to achieve relatively similar results to real personal data would affect data subjects' incentives to share their data in the first place. Such a study is beyond the scope of this paper.

⁷⁸ Rubinfeld & Gal, *supra* note 5, at 350-361.

⁷⁹ NVIDIA, *supra* note 37, at 8-9.

⁸⁰ Andrews, *supra* note 64 ("Because synthetic datasets are automatically labeled and can deliberately include rare but crucial corner cases, it's sometimes better than real-world data").

⁸¹ NIKOLENKO, *supra* note 20, at 6, 88.

⁸² Bryan McMahon, *AI is Ushering In a New Scientific Revolution*, THE GRADIENT (Jun. 4, 2022), <https://thegradients.pub/ai-scientific-revolution/>.

scientific breakthrough in the understanding and treatment of human disease.⁸³ Another example involves the use of synthetic data, based on various real sociodemographic conditions, for modelling micro-populations in order to evaluate the potential impact of different events.⁸⁴

Another reason synthetic data generation is more than a “cheap trick” is that, as noted above, synthetic data can *lead to new (or more accurate) learnings*. AlphaGo is a case in point. Or consider the following: some computer scientists are using synthetic data to fix highly accurate but overconfident AI models, the use of which is ill-suited for deployment in critical applications such as healthcare.⁸⁵ By adding synthetic data to create counterfactual explanations⁸⁶ for points that lie outside the model’s training distribution, or for novel cases that were not included in the collected data, they can test the accuracy of the model in such cases, potentially fixing its uncertainty characteristics while retaining its predictive value.⁸⁷ In a similar way, synthetic data can also be used to *enable new products or services*, such as creating virtual spaces in the metaverse.

Another part of the answer is that *machine learning algorithms can be trained on synthetic data* to increase their accuracy before using them in the real world.⁸⁸ For example, Nvidia uses synthetic data to train robots in warehouses to recognize objects of different shapes and sizes in different conditions, to make production lines more efficient.⁸⁹ A synthetic dataset was generated using artificial images, based on background knowledge incorporated into the generating algorithm about how lighting affects the appearance of images. The synthetic dataset was then used to train the robot.⁹⁰ In another prominent example, Amazon used synthetic data to train Alexa, its digital assistant, to apply voice recognition in three languages (Hindi, Spanish, and Brazilian Portuguese), for which it faced a shortage of collected data.⁹¹ Similar processes and rationales apply in the use of synthetic data to train autonomous vehicles under different road conditions, as discussed above.⁹² Indeed, acquiring and storing such data from live tests of real cars on real roads would be prohibitively expensive.⁹³ These examples also illustrate why we cannot always simply incorporate previous knowledge into a new algorithm without generating synthetic data: existing knowledge might be both too limited and too complex for efficient application. In the example above, the algorithm

⁸³ *Id.*

⁸⁴ NIKOLENKO, *supra* note 20, at 281.

⁸⁵ Sumedha Singla et al., *Augmentation by Counterfactual Explanation – Fixing an Overconfident Classifier* (unpublished manuscript, 2022), <https://arxiv.org/pdf/2210.12196.pdf>.

⁸⁶ Counterfactual explanations are a machine learning technique that describe causal relations in the form “If X had not occurred, Y would not have occurred.” They enable learning via consideration of hypothetical scenarios that contradict the observed facts. CHRISTOPH MOLNAR, *INTERPRETABLE MACHINE LEARNING: A GUIDE FOR MAKING BLACK BOX MODELS EXPLAINABLE* (2022), at Section 9.3.

⁸⁷ Sumedha et al., *supra* note 85, at 1.

⁸⁸ NIKOLENKO, *supra* note 20, at 13.

⁸⁹ Such services are being used by companies such as Amazon Robotics and PepsiCo. Andrews, *supra* note 64.

⁹⁰ NVIDIA, *supra* note 31, at 34.

⁹¹ Janet Slifka, *Tools for Generating Synthetic Data Helped Bootstrap Alexa's New-Language Releases*, AMAZON SCIENCE (October 11, 2019), <https://www.amazon.science/blog/tools-for-generating-synthetic-data-helped-bootstrap-alexa-s-new-language-releases>

⁹² Javier Tordable, *Synthetic Data Creates Real Results*, FORBES (August 26, 2020), <https://www.forbes.com/sites/googlecloud/2020/08/26/synthetic-data-creates-real-results/?sh=3c232b4b4d2f>

⁹³ *Id.*

operating the robot does not need to account for the myriad parameters describing how the same image looks under different lighting conditions. Rather, it needs to learn to react to the contours of *any* object under *any* lighting. For this purpose, synthetic images comprised a useful intermediate step in the algorithm’s learning.

Such uses build upon many of the comparative advantages of synthetic data noted above, including lower costs and increased speed of data acquisition and preparation; facilitating the creation of more representative datasets that include rare events and edge-case scenarios, thereby reducing bias in predictive models; and allowing for automatic and almost costless high-quality labelling without the need for manual annotation.⁹⁴ In addition, they allow seamless experimentation with different modalities.⁹⁵ For example, the performance of computer vision models is dependent on the quality of the sensor used to collect the training data, and how well it matches the sensor to be used in the final product (e.g., the quality of the lens).⁹⁶ Synthetic data that accommodates a wide variety of interchangeable sensor modalities limits the risks that might otherwise arise every time a product’s sensors are modified. It is thus not surprising that synthetic data has already become the main data resource to train machine learning systems, for experimentation and for prototyping of products and of algorithmic hypotheses.⁹⁷

Such training may also enable *faster learning*. To illustrate, the FDA is collaborating with researchers to explore the use of virtual patients in medical drug and device developments. Such patients can be duplicates of real medical profiles (“virtual twins”) which are used to test different, mock conditions in simulations. Alternatively, they can be completely virtual, computer-generated patients that represent the range of human variables,⁹⁸ used to replace or augment real patients. Such methods can potentially reduce human testing and shorten testing times. As one of the leading researchers in this area argues:

we should not limit ourselves by how the real world limits us. We can’t create a person that represents more than that person, but we can create a model that represents more than one person. Why not take advantage of that? Once you understand the diversity [of patients], you can build that into the [future, virtual] patient population.⁹⁹

Such tools are especially important where delays and costs impede patient access to novel treatments.¹⁰⁰ Of course, they may only be used where appropriate levels of safety and efficacy are ensured. This discussion also illustrates how synthetic data simulations can add value when it is *too risky to test different scenarios in the real world*.

⁹⁴ *Id.*

⁹⁵ *Id.*

⁹⁶ Shtylenko, *supra* note 15

⁹⁷ *Id.*

⁹⁸ Allison Proffitt, *The Role of Virtual Twins in Clinical Trials*, CLINICAL RESEARCH NEWS (Jul. 13, 2020), <https://www.clinicalresearchnews.com/news/2020/07/13/the-role-of-virtual-twins-in-clinical-trials> (Citing Steve Levine, the senior director of virtual human modeling at Dassault Systèmes).

⁹⁹ *Id.*

¹⁰⁰ *Id.*

Synthetic data can also be used to *ensure that learning is not focused on irrelevant, immoral, or illegal parameters*. A well-known example involves an algorithm that was trained to distinguish between images of Husky dogs and wild wolves. While the algorithm succeeded most of the time, the separating principle it adopted was insufficiently representative: it focused on the background, having learned that a white background (snow) signifies a wolf.¹⁰¹ To teach the algorithm not to focus only on the background, the coders applied a method called data perturbation, mixing and matching parts from different datasets. This process generated synthetic data in which pictures of wolves appeared against a range of backgrounds. The algorithm then learned to distinguish the two animals based on more relevant parameters. In another example, synthetic data was used to create a melanoma detection model that works on dark skin, where collected data repositories mostly contained images of lighter skin.¹⁰² Such benefits are not limited to imagery. One example is the introduction of artificial sentences to address bias in toxic sentence classification systems.¹⁰³

For the same reason, synthetic data can also assist analysts in *solving class imbalance problems* which arise when two groups of users are widely imbalanced, making it difficult to train algorithms on collected data.¹⁰⁴ Amazon's resume rating algorithm described above, which learned from its training dataset to discriminate against female applicants, offers a case in point. Another example involves JPMorgan's use of synthetic data to create an algorithm to detect money laundering. Given the scarcity of collected relevant data points, they used an automated simulator, based on examples of known behavior, to generate synthetic data points that provided a richer representation of the information a financial institution can observe when money laundering takes place. The coders then inserted the simulated data into a real dataset, according to predetermined probability distributions, and trained the algorithm to detect these instances.¹⁰⁵ Similar methods can also serve to *artificially overcome bias*: where a dataset correctly replicates existing societal prejudices and produces illegal discriminatory decisions that reduce social welfare,¹⁰⁶ synthetic data can be used to reflect norms of equality rather than real-world inequalities.¹⁰⁷

A final potential use of synthetic data is in *overcoming business constraints (such as trade secrets or data security) or legal ones (such as privacy regulation) on inter and intra-firm data transfers*.¹⁰⁸ That is, synthetic data can potentially serve as a "smokescreen" for sensitive

¹⁰¹ Marco Tulio Ribeiro, Sameer Singh & Carlos Guestrin, "Why Should I Trust You?" *Explaining the Predictions of Any Classifier*, in PROCEEDINGS OF THE 2016 CONFERENCE OF THE NORTH AMERICAN CHAPTER OF THE ASSOCIATION FOR COMPUTATIONAL LINGUISTICS: DEMONSTRATIONS 97 (2016).

¹⁰² Timo Kohlberger & Yuan Liu, *Generating Diverse Synthetic Medical Image Data for Training Machine Learning Models*, GOOGLE RESEARCH BLOG (Feb. 19, 2020), <https://ai.googleblog.com/2020/02/generating-diverse-synthetic-medical.html>.

¹⁰³ AGATHE BALAYN & SEDA GÜRSER, BEYOND DEBIASING: REGULATING AI AND ITS INEQUALITIES (2021), at fn 62.

¹⁰⁴ Caaritos Caritou, Simo Dragicevic, & Artur d'Avila Garcez, *Synthetic Data Generation for Fraud Detection Using GANs* (unpublished manuscript, 2021), <https://arxiv.org/pdf/2109.12546.pdf>

¹⁰⁵ S. Assefa et al., *Generating Synthetic Data in Finance: Opportunities, challenges and pitfalls*, in PROCEEDINGS OF THE 1ST INTERNATIONAL CONFERENCE ON AI IN FINANCE 2020 (2020).

¹⁰⁶ Barocas & Selbst, *supra* note 2, at 729-732.

¹⁰⁷ NIKOLENKO, *supra* note 20.

¹⁰⁸ Steven M. Bellovin, Preetam K. Dutta & Nathan Reitering, *Privacy and Synthetic Datasets*, 22 STANFORD TECH. L. REV. 1, 6 (2019) ("Synthetic data allows us to step away from the deidentification-reidentification arms race and focus on what really matters: useful data").

variables or for variables that are key identifiers, while preserving the (main) statistical attributes of the collected data.¹⁰⁹ This type of protection is used, for example, by the Federal Reserve and the Census Bureau in some of their published datasets.¹¹⁰ It is also widely used in healthcare, especially where the focus is on populations of patients, rather than individual patient records.¹¹¹ In addition, synthetic data can help firms meet updated privacy and security requirements while still making use of learnings from personal data which they were permitted to use in the past.

Of course, synthetic data also creates harms and risks. Most importantly, synthetic data models can be *inaccurate*.¹¹² This can result from incorrect or misrepresentative input data or background information. In particular, the coder can make erroneous assumptions about the distributions and correlations among the variables involved, which might arise when the process is new or the coder lacks sufficient experience with that system.¹¹³ This problem can be partially addressed by testing synthetic data against real-world outcomes, to determine and increase its accuracy. Perhaps more difficult to address is inaccuracy that results from data curation. Consider deidentification of collected data. Because privacy considerations imply that synthetic data cannot necessarily capture all the statistical relationships in the original data, such data manipulations can come at a cost: they create an inherent tradeoff between data protection and data utility (measured by the accuracy of the synthetic data as compared to collected data). However, although the field is still in its infancy, current methods for generating synthetic data already show promise in achieving high accuracy levels while lowering the risks of reverse engineering that would expose protected variables.¹¹⁴

In some situations, adding synthetic data increases the *risk of duplicating bias or errors*. Consider again Amazon's recruitment algorithm, which inadvertently perpetuated gender bias. Augmenting such a dataset with synthetic data could address this sort of bias by adding synthetic counter-examples (synthetic resumes of successful women). However, this sort of social engineering requires awareness and know-how. It is easy to imagine that adding synthetic data could perpetuate other kinds of biases of which coders are less aware.

Another potential concern with respect to synthetic data is that it may lead to complacency about the *risks of exposure* of the collected data used in its generation. While deidentification offers some protection, the continuing development of efficient algorithms, as well as synthetic datasets based on collected data, raises the prospect that

¹⁰⁹ J. P. Reiter, *Using CART to Generate Partially Synthetic Public Use Microdata*, 21(3) J. OF OFFICIAL STATISTICS 441 (2005); Bellovin, Dutta & Reitering, *Id.*, at 1.

¹¹⁰ Reiter, *Id.*

¹¹¹ See, e.g., *Synthetic Data*, CPRD (last visited January 28, 2023), <https://www.cprd.com/content/synthetic-data> (examples include a cardiovascular disease synthetic dataset and a COVID-19 symptoms and risk factors synthetic dataset).

¹¹² Theresa Stadler, Bristena Oprisanu & Carmela Troncoso, *Synthetic Data – Anonymisation Groundhog Day* (unpublished manuscript, 2022), at 1, <https://arxiv.org/pdf/2011.07018.pdf>.

¹¹³ This, however, does not necessarily imply that the synthetic data might not still be useful. A synthetic dataset based on inaccurate correlations can be used, for example, for debugging a data analysis program, or for some types of performance testing of software applications.

¹¹⁴ Vanessa Ayala-Rivera, Patrick McDonagh, Thomas Cerqueus & Liam Murphy, *Synthetic Data Generation using Generator Tool*, University College Dublin, Technical Report UCD-CSI-2013-03 (2018), <https://arxiv.org/pdf/1311.3312.pdf>. See also discussion *infra* Section III.A.

such datasets could be analyzed synergetically, leading to reidentification of data that would be impermeable to separate analysis. This is particularly true given advances in quantum computing, which can more easily break encryption methods.¹¹⁵

A final concern is that synthetic data could *change the power relationships* between players. By reducing barriers to data access, it can increase competition among suppliers/providers and facilitate data-based innovation. By potentially increasing the accuracy of the information held by suppliers/providers about consumers/users, it opens up more opportunities for beneficial use, but also for exploitation, manipulation, and abuse. This is true even if the supplier/provider does not possess significant market power, as long as he possesses relative bargaining power towards (some) consumers/users. The next section focuses on how synthetic data might affect power dynamics among suppliers/providers, while the following two focus on the effects on consumers/users.

III. EFFECTS ON COMPETITIVE DYNAMICS AND MARKET POWER

A. DATA-BASED MARKET POWER VIS-À-VIS COMPETITORS

Data-based advantages play an important role in the competitive dynamics of digital markets. This is because data is often the raw material for generation of information and knowledge, which enable better-informed decisions.¹¹⁶ The growth in machine learning applications, especially data-hungry deep learning techniques, is pushing the boundaries of what is economically feasible and physically possible.¹¹⁷ In such a setting, data-based advantages may not only strengthen the market power of some firms, but also make it more durable.¹¹⁸ As the OECD observed:¹¹⁹

[D]ata can give rise to self-perpetuating feedback loops, network effects and economies of scale that enhance the first-mover advantage of incumbent firms. Further, data access can be leveraged across multiple markets... Evidence suggests that market power may be on the rise, and that it may more durable, particularly in digital-intensive sectors.

Accordingly, competition in data and data-based markets is shaped by the height of

¹¹⁵ See, e.g., Tammy Xu, *What are Quantum-Resistant Algorithms—and Why Do We Need Them?*, MIT TECH. REV. (Sept. 14, 2022), <https://www.technologyreview.com/2022/09/14/1059400/explainer-quantum-resistant-algorithms/>.

¹¹⁶ Of course, this is not always the case. Take, for example, autonomous reinforcement learning models, such as the model used to train AlphaGo by DeepMind, which beat one of the top human players. AlphaGo needed almost zero training data, but much computational power. NIKOLENKO, *supra* note 20, at 10.

¹¹⁷ NIKOLENKO, *supra* note 20, at 1.

¹¹⁸ See, e.g., STIGLER REPORT, *supra* note 19, at 40.

¹¹⁹ OECD, COMPETITION COMMITTEE DISCUSSION PAPER, DATA PORTABILITY, INTEROPERABILITY AND DIGITAL PLATFORM COMPETITION 6 (2021), <https://www.oecd.org/daf/competition/data-portability-interoperability-and-digital-platform-competition-2021.pdf>. See also Frederic Jenny, *Competition Law Enforcement and Regulation for Digital Platforms and Ecosystems: Understanding the Issues, Facing the Challenges and Moving Forward* (unpublished manuscript, 2021), <https://ssrn.com/abstract=3857507>.

access barriers to data.¹²⁰ When such barriers are high, potential entrants might not be able to challenge incumbents who enjoy data-based advantages, because they cannot provide users with the utility that stems from better datasets.¹²¹ As a result, some data-based markets are characterized by limited contestability.¹²²

Such durable market power enables incumbents to enjoy high profit margins, which may lead to loss of allocative efficiency. But more importantly, productive and dynamic efficiency could be harmed. This is because firms with potential cost or quality advantages might not be able to enter the market, and the incentives of incumbents to develop consumer-welfare-enhancing innovations could be suppressed.¹²³ These negative welfare effects are strengthened by the lack of any guarantee that products offered by incumbents are the best of their kind. Rather, first-mover advantages can lock a market into a suboptimal technological equilibrium.¹²⁴ Furthermore, in recent years vertically integrated or conglomerate business models have become more commonplace in the digital marketplace, leading to the formation of data-based ecosystems – further raising entry barriers.¹²⁵

Synthetic data can help change such market dynamics. By introducing an alternative to some types of collected data, or by lowering the amounts of collected data needed, it can potentially *reduce obstacles in any part of the data value chain*.¹²⁶ Furthermore, given that synthetic data can be used to augment collected datasets which are otherwise too small to be useful,¹²⁷ firms with small datasets could compete with firms that possess much more collected data. Furthermore, in some industries synthetic data could even reduce the benefits of indirect network effects¹²⁸ for suppliers, and the resulting market structure of data-based ecosystems. Where, for example, synthetic data reduces the marginal benefit to a supplier from aggregating collected data from different sources (e.g., Facebook and Instagram), the comparative advantages firms can gain from such aggregations are reduced. This, in turn, could lead to more competition, more consumer choice (in both specific products and product bundles), and less concentrated market structures.

Synthetic data can also *change competitive dynamics via its effects on data sharing*. Where collected data no longer confers a significant comparative advantage on the collector,

¹²⁰ See OECD, DATA-DRIVEN INNOVATION: BIG DATA FOR GROWTH AND WELL-BEING (2015) (describing how data now drives all aspects of innovation in the economy and society), at 391–92; MAURICE E. STUCKE & ALLEN P. GRUNES, BIG DATA AND COMPETITION POLICY 79 (2016).

¹²¹ See STIGLER REPORT, *supra* note 19, at 40.

¹²² *Id.*, at 9, 34.

¹²³ *Id.*, at 8 (“[M]arket power may manifest itself through lower quality, lower privacy protection ...less variety of political viewpoints, and, importantly, less investments in innovation.”).

¹²⁴ See generally W. Brian Arthur, *Competing Technologies, Increasing Returns, and Lock-In by Historical Events*, 99 ECON. J. 116 (1989); W. BRIAN ARTHUR, INCREASING RETURNS AND PATH DEPENDENCE IN THE ECONOMY, 13–5 (1994).

¹²⁵ OECD, DATA PORTABILITY, *supra* note 119, at 5.

¹²⁶ For such obstacles see Rubinfeld & Gal, *supra* note 5, at 350–368.

¹²⁷ NIKOLENKO, *supra* note 20, at 12; Jim Hare, Svetlana Sicular & Erick Brethenoux, *Tech Providers 2025: Why Small Data Is the Future of AI*, GARTNER (Sept. 17, 2020), <https://www.gartner.com/en/documents/3990260/tech-providers-2025-why-small-data-is-the-future-of-ai>.

¹²⁸ Network effects which are mediated by a complement to the network, such as when a large number of sellers on a platform increases the number of consumers interested in using this platform. STIGLER REPORT, *supra* note 19, at 38.

his willingness to share it is increased. The market price of such data will be capped by the costs of generating comparable synthetic data. The ability to share data will also rise. To illustrate, should synthetic data – based on collected private data – not fall within the scope of privacy laws,¹²⁹ it need not comply with legal requirements that relate to data privacy (e.g., obtaining the consent of data subjects; developing and applying internal systems to limit the exposure of private data; or storing private data only in the jurisdiction where data subjects reside).¹³⁰ This, in turn, makes it easier to share such data, both internally and externally.¹³¹ While synthetic data might dampen firms' motivation to collect collected data, thereby obviously reducing the ability to share it, this dynamic may reduce wasteful collection of unneeded private data. Similar dynamics to those analyzed in this paragraph also apply to the sharing of synthetic data, once created.

Synthetic data may also potentially *reduce the collective action problem in data markets*, especially where it can be used to supplement and expand existing datasets. As Heller recognized in his seminal article "The Tragedy of the Anticommons," a breakdown in coordination may prevent the emergence of a commons even when general access to resources or infrastructure would be a social good.¹³² For example, Heller and Eisenberg showed that in the biomedical context, a patent thicket over otherwise synergetic pieces of information can stifle the creation of life-saving innovations that build on such information.¹³³ This dynamic is, of course, not limited to medical information. Rather, data collection dynamics involving transaction costs (such as harms to privacy) and strategic behavior (such as free riding) can affect the incentives of data subjects to provide their data, even when they know it will be put to good use which might benefit themselves as well as others.¹³⁴ Consider, for example, a situation where each data subject assumes that the data of others will have the same effect on the production of a good from which she can eventually benefit. If parting with such data involves even a small potential loss of privacy, she might not provide it. If many data subjects act in the same fashion, the collective good will not be created, reducing social welfare. By potentially lowering the number of collected data entries needed to make an informed decision, synthetic data can indirectly overcome this collective action problem.

Accordingly, the increased possibilities for internal data generation and internal and external data sharing may facilitate cumulative and synergetic knowledge production, which may stimulate competition and *generate new and better products or services*.¹³⁵ Most fundamentally, access to data can shape both the level and direction of

¹²⁹ See *infra* Section IV.

¹³⁰ Eur. Comm'n, Regulation (EU) 2016/679, General Data Protection Regulation of 2018 ("GDPR"), art. 46; Rubinfeld & Gal, *supra* note 5, at 364.

¹³¹ Harvard Business Review, *supra* note 30, at 6.

¹³² Michael A Heller, *The Tragedy of the Anticommons: Property in the Transition from Marx to Markets* 111(3) HARV. L. REV. 621 (1998).

¹³³ Michael A. Heller and Rebecca Eisenberg, *Can Patents Deter Innovation? The Anticommons in Biomedical Research*, 280 (5364) SCIENCE 698 (1998).

¹³⁴ Michael J. Madison, Brett M. Frischmann, Madelyn Sanfilippo & Katherine J. Strandburg, *Too Much of a Good Thing? A Governing Knowledge Commons Review of Abundance in Context*, 7 FRONTIERS IN RESEARCH METRICS AND ANALYTICS 1, 4-6 (2022).

¹³⁵ This was recognized by the European Commission: Eur. Comm'n, *Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions: A Digital Single Market Strategy for Europe* 14–15, COM (2015) 192 final (May 6, 2015). See also Harvard Business

innovative activity,¹³⁶ by increasing the diversity (of competitors and inputs) that is essential to creativity.¹³⁷ Furthermore, by moving comparative advantages away from data collection, synthetic data may encourage greater investment in data analysis and innovative applications to improve knowledge. Importantly, the positive effects of better knowledge may extend beyond the market for which the data are immediately relevant, due to transfer learning. The potential benefits of widening the use of most types of data are numerous.

Overcoming data-based comparative advantages is important not only for strengthening competition within a jurisdiction, but also for *overcoming comparative advantages of other jurisdictions*. To illustrate, consider the comparative advantage the Chinese government created for Alibaba by enabling it to test its algorithms for smart cities in several locations.¹³⁸ It would be very difficult for any other firm to accumulate so much collected data, especially given privacy concerns. If, however, this can be partly overcome by a synthetic dataset, then synthetic data can increase competition.

Of course, synthetic data does not necessarily reduce all data access barriers. Access to collected data is still needed where the creation or utility of synthetic data are based on collected data, or where access to collected data can make the production of synthetic data cheaper. For example, Meta recently bought the image rights to all cricket matches in India,¹³⁹ giving it a large image bank on which to train its algorithm to create synthetic images of cricket players for the metaverse. A competing firm might buy a similar image bank for another sport. Yet where access to the necessary collected data is characterized by high barriers, control of such data could strengthen a firm's market power, for two main reasons. First, its potential use to create more and better synthetic data increases the comparative advantages it offers. In that sense, less is more. Second, by reducing obstacles to internal data flows, including the costs of adopting data governance systems that comply with legal data privacy and security requirements, synthetic data enables (better) enjoyment of internal data-based economies of scale and scope by those who already possess large quantities of collected data. To illustrate, consider the case of JPMorgan, where data anonymization through the generation of a synthetic dataset, based on private data, enabled it to share data on risk analysis across its departments.¹⁴⁰ While such increased internal data flows raise a firm's utility from the data, they could also strengthen the market power of digital ecosystems, increasing both horizontal and vertical concentration, and heightening entry barriers.

Where the generation of synthetic data requires specific expertise, including

Review, *supra* note 30, at 6.

¹³⁶ Jeffrey L. Furman & Scott Stern, *Climbing atop the Shoulders of Giants: The Impact of Institutions on Cumulative Research*, 101 AM. ECON. REV. 1933, 1936 (2011); Heidi L. Williams, *Intellectual Property Rights and Innovation: Evidence from the Human Genome*, 121 J. POL. ECON. 1, 1–2 (2013).

¹³⁷ Wolfgang Kerber & Simonetta Vezzoso, *Dow/Dupont: Another Step Towards a Proper Assessment Concept of Innovation Effects of Mergers* (unpublished manuscript, 2019), <https://ssrn.com/abstract=3856885>.

¹³⁸ Tamar Giladi Shtub & Michal S. Gal, *The Competitive Effects of China's Legal Data Regime*, 18(4) J. COMP. L. AND ECON. 936 (2022).

¹³⁹ Manish Singh, *Facebook secures exclusive digital rights for ICC cricket events*, TECHCRUNCH (Sept. 26, 2019), <https://techcrunch.com/2019/09/26/facebook-secures-exclusive-digital-rights-to-stream-icc-global-events-in-indian-sub-continent/?guccounter=1>.

¹⁴⁰ JPMorgan, *supra* note 35.

methods based on protected intellectual property, this could create another entry barrier. However, projects like the Synthetic Data Vault, released in 2020 by MIT researchers,¹⁴¹ reduce such obstacles by creating external synthetic data generators. The project uses synthesizers which allow users to upload their data and receive a new dataset with the same statistical properties as their original data, which can be made public without infringing privacy.¹⁴² Firms may also outsource the creation of synthetic data, making it affordable at smaller scales. Finally, open-source communities, such as Open Synthetics, develop and offer the use of some synthetic data generators.¹⁴³

B. IMPLICATIONS FOR ANTITRUST AND REGULATION OF PLATFORMS

Competitive dynamics form an important basis for the application of laws designed to ensure that, where possible, competition can take its course, so that consumers can enjoy what markets have to offer. Such laws include, *inter alia*, antitrust and platform regulations. They are based on assumptions regarding the operation of markets, which determine when regulation is justified. Given that synthetic data can potentially affect the competitive dynamics in data-based markets, it may change the applicability of such laws. Here we explore several examples.

Let us first focus on antitrust, which aims to prevent the erection of artificial barriers to competition. *Market power* is a foundational prerequisite for most antitrust prohibitions, including monopolization and merger regulation. This is because absent market power, competitive concerns are limited.¹⁴⁴ It is well-established that collected data can create significant market power in some digital markets.¹⁴⁵

Synthetic data may affect findings of market power. To illustrate, suppose a newcomer wishes to compete in the market for autonomous trains. To do so, he needs numerous videos of images seen from the train. Such videos are generally the proprietary information of incumbents. But if he can generate such images synthetically, by using a small number of real images, this would significantly reduce entry barriers. Put differently, synthetic data may break down economies of scale and scope in the use of collected data in some markets.

These potential changes in data-based market power have implications for all parts of antitrust law. Below we focus on two types of effects: instances in which the possible generation of synthetic data affects the *need* for regulatory intervention in the marketplace, and those in which it affects *findings* of anti-competitive conduct. While

¹⁴¹ MIT Laboratory for Information and Decision Systems, *The Real Promise of Synthetic Data: MIT Researchers Release the Synthetic Data Vault, a set of Open-Source Tools Meant to Expand Data Access Without Compromising Privacy*, MIT NEWS (Oct. 16, 2020), <https://news.mit.edu/2020/real-promise-synthetic-data-1016>

¹⁴² The Synthetic Data Vault, <https://sdv.dev/SDV/> (last visited January 23, 2023).

¹⁴³ See, e.g., Open Synthetics, *The Open Community for the Creation and Use of Synthetic Data in AI*, <https://opensynthetics.com/> (last visited January 23, 2023).

¹⁴⁴ For an argument that challenges this foundational principle see generally Blickstein-Shchory & Michal S. Gal, *Market Power Parasites*, 35(1) HARV. J. L. TECH. 73 (2021).

¹⁴⁵ See, e.g., Filippo Lancieri and Patricia Sakowski, *Competition in Digital Markets: A Review of Expert Reports*, 26 STAN. J.L. BUS. & FIN. 65 (2021). It is also suggested that data can give rise to forms of power not neatly captured by the concept of market power, see, e.g., Orla Lynskey, *Grappling with "Data Power": Normative Nudges from Data Protection and Privacy*, 20(1) THEORETICAL INQUIRIES IN LAW 189 (2019) and, more broadly, JULIE COHEN, *BETWEEN TRUTH AND POWER: THE LEGAL CONSTRUCTIONS OF INFORMATIONAL CAPITALISM* (2019).

such effects will generally be case-specific, in some industries they may be generalized.

Let us start with several examples of the first effect. Take the regulation of *mergers and acquisitions*. The digital era is characterized by high-profile mega-mergers that involve vast amounts of data relating to user conduct on digital platforms. These include, inter alia, Google/DoubleClick, Microsoft/Yahoo!, Microsoft/Skype, Microsoft/LinkedIn, Facebook/Instagram, Facebook/WhatsApp, and Google/Fitbit.¹⁴⁶ Strong criticisms have been voiced against the clearance of such data-based mergers by antitrust authorities.¹⁴⁷ Enter synthetic data. Where data-based comparative advantages can be (partly) overcome, the negative welfare effects of changes in market structure will be lower. Accordingly, in industries where firms will be able to compete with smaller quantities of collected data, and where the collection of such data does not involve insurmountable barriers, more mergers would be benign.

At the same time, as not all data could be created artificially, some data-based mergers should still be prohibited. This can be exemplified by the case brought against Facebook by the Federal Trade Commission in December 2020.¹⁴⁸ One of the main allegations concerned Facebook's acquisition of the social networking applications WhatsApp and Instagram: These acquisitions strengthened Facebook's market power, on the grounds that, inter alia, "over time, users of a personal social network build more connections and develop a history of posts and shared experiences, which they would lose by switching to another personal social networking provider."¹⁴⁹ Such data could not be artificially re-created, at least not easily.¹⁵⁰ Furthermore, when synthetic data strengthens comparative advantages, as in the case of JPMorgan elaborated above, the need for merger review is increased.

Moreover, the application of antitrust should also be sensitive to the effects of internal transfers of deidentified data. Consider the following example, offered by Stavroulaki:¹⁵¹ The Affordable Care Act prevents health insurers from discriminating against citizens based on their pre-existing conditions.¹⁵² One way to circumvent this prohibition is to offer less coverage in geographic areas where consumers are more prone to suffer from certain medical conditions. The data required for such a strategy can be based on patients' medical records. While such records are protected under

¹⁴⁶ Anca D. Chirita, *Data-Driven Mergers Under EU Competition Law*, in *THE FUTURE OF COMMERCIAL LAW: WAYS FORWARD FOR HARMONISATION*, 147 (J. Linarelli & O. Akseli eds., 2019); Jörg Hoffmann and Germán Oscar Johannsen, *EU-Merger Control & Big Data: On Data-specific Theories of Harm and Remedies*, in *EU COMPETITION LAW REMEDIES IN THE DATA ECONOMY* (Marco Botta & Josef Drexler eds., 2019).

¹⁴⁷ See, e.g., Chirita, *Id.*, at 147 ("No merger should be unconditionally cleared if it involves a large amount of users' data").

¹⁴⁸ Complaint, Case 1:20-cv-03590, *FTC v. Facebook Inc.* (Dec. 9, 2020).

¹⁴⁹ *Id.*, para. 66.

¹⁵⁰ For an additional example see, e.g., the Ticketmaster/ Live Nation merger that was conditionally approved in January 2010 (data on ticketing data (including data on number of tickets sold, proceeds from those sales, ticket inventory, pricing, marketing and corresponding sales) and ticket buyer data (including nonpublic identifying information for ticket buyers) could not be artificially created). Judgment, *U.S., et al. v. Ticketmaster Entertainment, Inc., et al.*, No. 10-cv-00139 (D.D.C. 30 July 2010); Amended Final Judgment, *U.S., et al. v. Ticketmaster Entertainment, Inc., and Live Nation Entertainment, Inc.*, No. 1:10-cv-00139 (D.D.C. 28 January 2020).

¹⁵¹ Theodosia Stavroulaki, *How the Wrong Presumptions Led to the Wrong Conclusions in the United/Change Healthcare Merger*, PRO MARKET (Nov. 11, 2022).

¹⁵² Patient Protection and Affordable Care Act, Pub. L. No. 111-148, 124 Stat. 119 (2010).

privacy laws, their transfer or sale is allowed if the data is deidentified. Such strategic use of data could result, it is argued, from the recent merger between UnitedHealth Group, a giant health insurer, and Change Healthcare, the largest U.S. electronic data interchange clearinghouse.¹⁵³ This is because the approved merger will give UnitedHealth access to deidentified health data on millions of patients, enabling it to cherry-pick the most profitable geographic areas. This, in turn, would give UnitedHealth a competitive advantage, especially if its rivals are deprived of access to a similar type and range of data.¹⁵⁴

Synthetic data can also *reduce the need and justification for data sharing arrangements* in order to realize data synergies. If such synergies can be realized via internal generation of synthetic data, then such arrangements, especially between competitors, have weaker justification. This observation has implications for many aspects of antitrust, as well as for ex-ante regulation of the digital economy which seeks to advance more efficient use of data.

Take, for example, *mandatory data access*. Given the importance of data for the efficient operation of many markets, several laws mandate data sharing, including the Essential Facilities Doctrine in antitrust.¹⁵⁵ The doctrine mandates sharing on fair and non-discriminatory terms of facilities whose use is essential for competition in an adjacent market.¹⁵⁶ While the wings of the doctrine were clipped in the Supreme Court's *Trinko* decision,¹⁵⁷ the digital economy has revived calls for its application to data which is essential for competition. Graef, for example, suggests recognizing some types of data as "essential data".¹⁵⁸ She is not alone.¹⁵⁹ Such calls advance the claim that, despite data's non-rivalrous nature, some forms of data are indispensable, given that obstacles make it impossible or extraordinarily difficult to collect or to re-create it.¹⁶⁰ Synthetic data may challenge such essentiality in some markets: if collected data can be replaced by synthetic data, then the justifications for requiring firms to share it are weakened. This does not

¹⁵³ Stavroulaki, *supra* note 151.

¹⁵⁴ *Id.*

¹⁵⁵ MCI Commc'ns Corp. v. AT&T, 708 F.2d 1081, 1132-33 (7th Cir. 1983) (Terms for application include: control of the essential facility by a monopolist; a competitor's inability practically or reasonably to duplicate the essential facility; denial of the use of the facility to a competitor; and the feasibility of providing the facility).

¹⁵⁶ *Id.*

¹⁵⁷ Verizon Commc'ns Inc. v. Law Offices of Curtis V. Trinko, 540 U.S. 398, 411 (2004) (interpreting the duty to deal articulated in *Aspen Skiing* as confined to a setting in which "[t]he unilateral termination of a voluntary (and thus presumably profitable) course of dealing suggested a willingness to forsake short-term profits to achieve an anticompetitive end.") The doctrine has been criticized based on concerns about incentives for dynamic innovation, trust in the self-correcting mechanisms of markets, denials of the very existence of incentives to monopolize adjacent markets, and dire assessments of the ability of the courts and agencies to replace market mechanisms took center stage. See generally Zachary Abrahamson, *Essential Data*, 124 YALE L.J. 867 (2014).

¹⁵⁸ INGE GRAEF, EU COMPETITION LAW, DATA PROTECTION AND ONLINE PLATFORMS: DATA AS ESSENTIAL FACILITY (2016). In some rare cases an obligation to share data has been imposed through antitrust laws in order to advance competition. See, e.g., Vikas Kathuria & Jure Globocnik, *Exclusionary Conduct in Data-Driven Markets: Limitations of Data Sharing Remedy*, 8 J. ANTITRUST J. ENFT 511, 519-20 (2020).

¹⁵⁹ EU Report, *supra* note 1 ("The 'classical' EFD may not be the right framework to handle refusal of access to data cases, as the doctrine has been developed with a view to access to 'classical' infrastructures and later expanded to essential IPRs.").

¹⁶⁰ For criticism and highlights of the difficulties in applying the doctrine to data, see, e.g., Giuseppe Colangelo, & Mariateresa Maggolino, *Big Data as Misleading Facilities*, 13 (2) EUR. COMP. J. 249, 249-50 (2017).

imply, of course, that no dataset will ever be deemed indispensable, but it may significantly reduce the number of such instances. The same logic applies to mandatory data sharing as a remedy for anti-competitive conduct.¹⁶¹ Sharing will be necessary for reintroducing competition only if there are no alternative cost-effective ways of duplicating ill-gotten data-based advantages. At the same time, where sharing can assist in reintroducing competition, the ability to deidentify private data via synthetic data increases such a possibility.

Synthetic data also affects the need for *mandated data portability, interoperability, and standardization*. All three involve technical standards that affect the ability to learn from the data, and are often mentioned as key elements of a digital competition policy reform agenda.¹⁶² Data portability seeks to reduce user switching costs, for example by mandating that he be able to transfer his personal data to another provider. Interoperability focuses on allowing systems to communicate with one another, increasing the ability of digital services to incorporate data or functionality from another data provider.¹⁶³ Data standardization ensures that data is understandable by the receiver and is in usable format.¹⁶⁴ All three measures have already been implemented through antitrust enforcement, sector-specific regulation, and other broad-based regulations around the world.¹⁶⁵ For example, data portability in banking was incorporated through the Dodd Frank Act, and several new interoperability measures have been proposed for the digital sector.¹⁶⁶ Synthetic data reduces the need for such policy tools in some markets. This is important given that such measures carry their own costs.¹⁶⁷ A recent OECD study emphasized that developing legal, technical, and procedural aspects of these measures may be particularly complex, as will monitoring.¹⁶⁸ Such policy tools might also inadvertently create negative effects. For example, data standardization can entrench inefficient collection standards.¹⁶⁹ Accordingly, synthetic data could reduce the need to use second-best solutions for expanding data use.

Let us now turn to the second type of effect, which *changes the content of antitrust prohibitions*. Synthetic data may change, for instance, optimal burdens of proof. Take, for example, a case where a monopolist limits the data portability or interoperability associated with its product. Determining whether such a change constitutes an illegal refusal to deal is based, *inter alia*, on the assessment of harm to competition.¹⁷⁰ Should synthetic data generally limit the need for interoperability, harm to competition will be reduced, and a higher burden of proof might be justified across all cases.

¹⁶¹ See generally Kathuria & Globocnik, *supra* note 158.

¹⁶² OECD, *supra* note 119, at 6, 22 (For providers of complementary services, linkages with a central ecosystem platform may also be its primary means of attracting users, and a way of leveraging existing functionality, such as account authentication and sign-in functions. If an incumbent makes small changes to APIs or layers on additional procedures, it could have a fatal effect on the business model of firms relying on the API).

¹⁶³ *Id.*, at 9

¹⁶⁴ Michal S. Gal & Daniel M. Rubinfeld, *Data Standardization*, 94 NYU L. REV. 737, 749 (2019).

¹⁶⁵ OECD, *supra* note 119, at 7, 24.

¹⁶⁶ *Id.*

¹⁶⁷ *Id.*, at 15.

¹⁶⁸ *Id.*

¹⁶⁹ *Id.*, at 11; Rubinfeld & Gal, *supra* note 5, at 762.

¹⁷⁰ OECD, *id.*

Another example relates to the substance of laws. To illustrate, consider the prohibition of *cartels*, which is based on the existence of an “agreement in restraint of trade”. Synthetic data on market conditions and rival's actions can help train algorithms to reach a coordinated equilibrium without such an agreement,¹⁷¹ and the algorithm can then be applied to real-world conditions. In other words, synthetic data can be used to circumvent the existing law, in a way that can only be addressed by reformulating the content of the prohibition.¹⁷²

Similar considerations to those explored in this section should affect a host of additional regulatory tools which are grounded in the assumption that data-based advantages are significant, and cannot be easily overcome. Take, for example, the European Digital Markets Act,¹⁷³ one of the first regulations in the world geared specifically towards digital platforms, which also applies to U.S.-based firms in their dealings in the EU.¹⁷⁴ Some of its prohibitions were designed to limit comparative advantages based on collected data. For example, it mandates the gatekeeper platform to provide competing providers of online search engines with access on fair, reasonable, and non-discriminatory terms to ranking, query, click, and view data generated by searches on its engines, subject to anonymization of personal data.¹⁷⁵ This regulation, which makes use of synthetic data to enable data transfers, is based on the assumption that such data creates significant data-based comparative advantages that could entrench the market power of the platform. A similar assumption stands at the basis of some recent U.S. legislative proposals. To illustrate, the Augmenting Compatibility and Competition by Enabling Service Switching (ACCESS) Act would mandate dominant platforms to maintain certain standards of data portability and interoperability.¹⁷⁶ It might be better, however, to limit such regulations to those types of data in which data-based advantages cannot easily be overcome once the option of synthetic data is introduced.

Many of the examples explored above suggest that the introduction of synthetic data will lead to a less interventionist approach to data-based advantages that affect competition, especially those that result from the natural conditions of the market rather than anti-competitive conduct. At the same time, antitrust and platform regulation have an important role to play in ensuring that artificial barriers to the creation of synthetic data are not erected, and that where synthetic data increases market power, such power is not abused.

¹⁷¹ Emilio Calvano, Giacomo Calzolari, Vincenzo Denicolò, & Sergio Pastorello, *Artificial Intelligence, Algorithmic Pricing, and Collusion*, 110 AM. ECON. REV. 3267 (2020).

¹⁷² See, e.g., ARIEL EZRACHI & MAURICE E. STUCKE, VIRTUAL COMPETITION: THE PROMISE AND PERILS OF THE ALGORITHM-DRIVEN ECONOMY (2016); Michal S. Gal, *Algorithms as Illegal Agreements*, 34 BERKELEY TECH. L. J. 67, 84-87 (2019).

¹⁷³ Regulation (EU) 2022/1925 of the European Parliament and of the Council of 14 September 2022 on Contestable and Fair Markets in the Digital Sector and Amending Directives (EU) 2019/1937 and (EU) 2020/1828, 2022 O.J. (L 297) 15 (“Digital Markets Act”).

¹⁷⁴ See generally Kevin Davis & Florencia and Marotta-Wurgler, *Contracting for Personal Data*, 94(4) NYU L. REV. 662 (2019).

¹⁷⁵ Digital Markets Act, *supra* note 173, Article 6(11).

¹⁷⁶ H.R.3849, 117th Cong. (2021). On May 25, 2022, the bill was referred to the House committees of jurisdiction.

Synthetic data alters not only the power relationships between competitors in data-driven markets, but also between those who use data for decision-making and those who are affected by such decision-making, whether individuals or groups. Indeed, synthetic data can *affect the bargaining power of those that possess data*, even if they do not possess market power in the antitrust sense. The next section reviews effects on privacy, while the following one focuses on the effects of increased data quality on users/consumers.

IV. EFFECTS ON DATA PRIVACY

Privacy laws govern the handling of personal information. At their essence, they constrain the power such information confers.¹⁷⁷ As such, they constitute essential vehicles for the protection of human rights, such as autonomy and self-definition, and act as facilitators of social institutions, including democracy and trust.¹⁷⁸ They seek to strike a social-welfare-enhancing balance between various interests, including fundamental human rights and data utility, both of which may benefit data holders, individuals, and society at large.¹⁷⁹ In this section we identify the impact of synthetic data on this balance, and examine the challenges of applying existing privacy laws to such data.

Technological change, in particular the increased possibility of reidentifying deidentified data, has already interrupted the calibration of interests promoted by privacy laws.¹⁸⁰ Synthetic data further disrupts this balance. Moreover, as we show, the lens of synthetic data illuminates many of the problems that exist with our current legal conceptions of privacy harms.

A. THE IMPACT OF SYNTHETIC DATA ON BALANCING OF INTERESTS IN PRIVACY LAWS

Common to all data privacy regimes, albeit to differing extents, are principles that promote the fairness of data processing.¹⁸¹ The requirement of lawful processing,

¹⁷⁷ See, e.g., ORLA LYNKEY, *THE FOUNDATIONS OF EU DATA PROTECTION LAW* (2015); DANIELLE CITRON, *THE FIGHT FOR PRIVACY PROTECTING DIGNITY, IDENTITY AND LOVE IN OUR DIGITAL AGE* (2022); NEIL RICHARDS, *WHY PRIVACY MATTERS* 39 (2022).

¹⁷⁸ Global Privacy Assembly, *Privacy and Data Protection as Fundamental Rights—A Narrative*, Report Prepared for the GPA WG3 on Privacy and Human Rights, 43 GPA PSWG3, 2022, <https://globalprivacyassembly.org/wp-content/uploads/2022/03/PSWG3-Privacy-and-data-protection-as-fundamental-rights-A-narrative-ENGLISH.pdf>.

¹⁷⁹ See generally Madison et al., *supra* note 134.

¹⁸⁰ See generally Paul Ohm, *Broken Promises of Privacy: Responding to the Surprising Failure of Anonymization* 57 UCLA L. REV. 1701 (2019); Omer Tene & Jules Polonetsky, *Big Data for All: Privacy and User Control in the Age of Analytics*, 11 NW. J. TECH. & INTELL. PROP. 239 (2013).

¹⁸¹ In the U.S. these are the so-called Fair Information Practice Principles (FIPPS). See, e.g., Woodrow Hartzog, *The Inadequate, Invaluable Fair Information Practices*, 76 MD. L. REV. 952, 959 (2017); LEE BYGRAVE, *DATA PRIVACY LAW: AN INTERNATIONAL PERSPECTIVE* 146 (2014).

frequently reflected in consent requirements, is well-known.¹⁸² Additional principles include, among others, requirements regarding data security, data minimization, and data quality.¹⁸³ Synthetic data can *potentially promote these latter principles*. Most importantly, by replacing collected data with artificially generated data, or by adding to a dataset synthetic data that screens the outlier data points while retaining the statistical properties of the data, synthetic data offers an additional layer of security to personal data.¹⁸⁴ Data minimization requires that only the minimum amount of personal data be processed for a specified purpose.¹⁸⁵ As synthetic data is generated on demand, often with a specific use envisaged, it reduces the incentives for data holders to gather excess data.¹⁸⁶ Finally, by augmenting datasets where collected data is unavailable, synthetic data can promote data accuracy.¹⁸⁷ In light of these advantages, data privacy regulators have recommended the use of synthetic data as a substitute for collected data in certain contexts. For example, the Norwegian Confederation of Sports unintentionally shared the personal data of 3.2 million Norwegians online when testing solutions for moving a database from a physical server to the cloud. The Norwegian data privacy regulator emphasized that this could have been avoided had synthetic data been used to test the migration.¹⁸⁸

Moreover, some computer scientists argue that synthetic data can, in some situations, *improve the privacy–utility tradeoff relative to other privacy-enhancing techniques*.¹⁸⁹ Utility requires retaining in the dataset the relationship between different features and the distribution of values for each feature.¹⁹⁰ An optimal balance would preserve privacy while also allowing the data to be analyzed for socially valuable ends. For instance, in the medical context, it should be possible to process patient information so as to protect privacy while still enabling providers to mine the data for insights into illnesses and new treatments. Many existing techniques achieve anonymization by introducing noise into a dataset, thus disturbing the relationship between attributes or the distribution of values for attributes, or by stripping the dataset of some meaningful data points.¹⁹¹ Such techniques reduce the accuracy, and therefore the utility, of data. Some computer scientists claim that synthetic data may significantly recalibrate this inverse relationship,

¹⁸² Hartzog, *id.*

¹⁸³ *Id.*

¹⁸⁴ We follow Maynard-Atem, who considers synthetic data to be “a subset of anonymisation created by an automated process such that it holds similar statistical patterns as the original dataset.” Louise Maynard-Atem, *The Data Series – Solving the Data Privacy Problem Using Synthetic Data*, 2019(2) IMPACT 11 (2019).

¹⁸⁵ See, e.g., Hartzog, *supra* note 181, at 957 (the principle is called there the Collection Limitation Principle”).

¹⁸⁶ Note, however, that this does not require, or imply, that the overall amounts of data produced will be lower.

¹⁸⁷ See *infra* Section V.

¹⁸⁸ European Data Protection Board, *Norwegian DPA: Norwegian Confederation of Sport fined for inadequate testing*, EDPB (June 15, 2021), https://edpb.europa.eu/news/national-news/2021/norwegian-dpa-norwegian-confederation-sport-fined-inadequate-testing_en.

¹⁸⁹ See discussion in this para.

¹⁹⁰ Features or attributes are the characteristics of an object studied (e.g., age; height), while variables are the values assigned to these attributes (e.g., young/old; 18-35).

¹⁹¹ See, e.g., EU ARTICLE 29 DATA PROTECTION WORKING PARTY, OPINION 05/2014 ON ANONYMIZATION TECHNIQUES (2014) (Discussing noise addition, permutation, differential privacy, aggregation, k-anonymity, l-diversity and t-closeness). A common challenge for state-of-the-art methods such as differential privacy is mainly on how to balance the added noise with the utility of the data. More noise usually means less privacy risk, but also means less utility on the data. Zhenchen Wang, Puja Myles & Allan Tucker, *Generating and Evaluating Cross-Sectional Synthetic Electronic Healthcare Data: Preserving Data Utility and Patient Privacy*, 37 COMPUTATIONAL INTELLIGENCE 819, 825 (2021).

by “maintain[ing] the majority of the valuable information and statistical integrity of the original data while reducing the risk of re-identification”.¹⁹² Accordingly, it can potentially act as a relatively accurate proxy for collected data, but with lower privacy risks. Notably, however, privacy gains might not be evenly distributed: as discussed below, synthetic data may offer better privacy protection for some records.¹⁹³ Some individuals might therefore be disadvantaged relative to others. Still, the result may be Pareto-optimal relative to the state without synthetic data. As elaborated below, some synthetic data proponents make an even stronger claim: that by combining synthetic data with another privacy-preserving technique, namely differential privacy, the resulting dataset retains both privacy and utility.¹⁹⁴

Even if this is the case, and we do not take a stance on this point, an interpretability challenge may ensue.¹⁹⁵ A machine learning model is interpretable if “you can inspect the actual model and understand why it got a particular answer for a given input, and how the answer would change when the input changes”.¹⁹⁶ Put differently, interpretability involves the ability to understand the algorithm’s functioning and its outputs, rather than providing an answer to why the specific model was chosen.¹⁹⁷ When differential privacy and synthetic data are combined, it is not possible to know which patterns of the original dataset are retained and which are lost. This has led some to conclude that “synthetic data leads to a highly variable privacy gain and unpredictable utility loss”.¹⁹⁸ Yet, as explained in Section II above, both parameters can be tested against the statistical properties of the collected data. We analyze the consequences of this reduction of interpretability further below. However, for now suffice to note that synthetic data might introduce another cost into the equation.

Furthermore, where collected data is used to generate a synthetic dataset, there is always some risk that the model itself or the dataset might indirectly leak some of the original personal data.¹⁹⁹ The attacker would essentially be reverse engineering the

¹⁹² Maynard-Atem, *supra* note 184, at 12; F.K Dankar and M. Ibrahim, *Fake It Till You Make It: Guidelines for Effective Synthetic Data Generation*, 11(2158) APPL. SCI. 4 (2021); Bellovin, Dutta, and Reiter, *supra* note 108, at 4. For a different view see, e.g., T. Stadler, B. Oprisanu, & Carmela Troncoso, *Synthetic Data—Anonymisation Groundhog Day*, in 31ST USENIX SECURITY SYMPOSIUM, 1451 (2022).

¹⁹³ See *infra* sub-section B.

¹⁹⁴ Stadler et al., *supra* note 112. For an overview of some claims with regard to privacy see, e.g., Dankar & Ibrahim, *supra* note 192.

¹⁹⁵ Dara Hallinan & Frederik Zuiderveen Borgesius, *Opinions can be Incorrect (in our opinion)! On Data Protection Law’s Accuracy Principle*, 10(1) INT’L DATA PRIVACY L. 1, fn. 18 (2020) (“Given the lengthy use of the concepts of quality and accuracy in computer science, it is somewhat of a peculiarity that there has not been more cross-fertilization of ideas with the accuracy concept in data protection law”).

¹⁹⁶ STUART RUSSELL & PETER NORVIG, *ARTIFICIAL INTELLIGENCE: A MODERN APPROACH* 729 (4th ed., 2022). Interpretability (and explainability) is not always defined similarly by all researchers. See, e.g., Cynthia Rudin, *Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead*, 1 NATURE MACHINE INTELLIGENCE 206 (2019) (“an interpretable machine learning model is constrained in model form to be either useful to someone or to obey structural knowledge of the domain, such as monotonicity, causality, structural (generative) constraints, or physical constraints that come from domain knowledge.”).

¹⁹⁷ Christoph Molnar, *Interpretable Machine Learning*, GITHUB (Dec. 14, 2022), <https://christophm.github.io/interpretable-ml-book/>.

¹⁹⁸ Stadler et al, *supra* note 112, at 1.

¹⁹⁹ An empirical study conducted by Stadler et al. found that synthetic data does not always protect the original real records from linkage and attribute inference. *Id.* See also Khaled El Emam, Lucy Mosquera, &

collected data from the model. Most notably, synthetic data that attempts to retain all the statistical properties of the collected data also retains the risk of linking the data to a specific person in outlier situations. Writing in the context of medical data, Wang et al. warn that the reidentification of outliers in the dataset could be an issue in cases where the synthetic dataset is “very similar in terms of aggregated characteristics to real-world data”.²⁰⁰ This problem is also recognized in the industry literature. As a Gartner executive observes, if “you are creating data for a rural area and it’s one person per 100 miles, even though I can create a synthetic person, it doesn’t hide anything”.²⁰¹ Solving the outlier risk problem often requires a reduction in the utility of the data.²⁰² Note, though, that in some situations it is possible to add to the dataset synthetic data which conceals the association of the data to a specific person by enlarging the group of data subjects which exhibit the outlier features. Thus, while synthetic data can potentially significantly reduce the risks of reidentification, it is not always the game-changer for the privacy–utility tradeoff that some suggest. This also points to the dangers of treating synthetic data as distinct from other anonymization techniques.

More fundamentally, synthetic data could create privacy harms even when a direct link does not exist between the individual and the personal data protected. This direct link, which is required in most privacy laws, is challenged in the context of horizontal data relations, where data about individual A (or individuals in group A) is used in relation to, or to learn about, individual B (or individuals in group B). While it has already been recognized that modern data analytics techniques put pressure on this relational dimension of privacy laws with regard to collected data,²⁰³ as elaborated below synthetic data places further strain on their capacity to tackle data externalities.

Synthetic data therefore accentuates existing challenges to the effectiveness of data privacy laws, calling into question their rationale and the assumptions in which they are grounded. With this in mind, we focus on two main questions. We ask, doctrinally, whether synthetic data is captured under privacy laws; and normatively, whether existing laws are fit to protect against privacy harms, while not harming data flows to an unnecessary extent, in a world in which synthetic data is widely used.

B. APPLICATION OF PRIVACY LAWS TO SYNTHETIC DATA

Is synthetic data captured under privacy laws?²⁰⁴ Anonymous data escapes the application of data privacy frameworks worldwide.²⁰⁵ This has led to some categorical

Jason Bass, *Evaluating Identity Disclosure Risk in Fully Synthetic Health Data: Model Development and Validation*, 11 J. MEDICAL INTERNET RESEARCH 22, 22 (2020).

²⁰⁰ Wang, Myles & Tucker, *supra* note 191, at 821.

²⁰¹ Cited in Harvard Business Review, *supra* note 31, at 4.

²⁰² Wang, Myles & Tucker, *supra* note 191, at 826.

²⁰³ See, e.g., Christopher Jon Sprigman & Stephan Tontrup, *The Paradox of Privacy Regulation* (2022)(on file with author).

²⁰⁴ See generally Bellovin et al., *supra* note 108; Cesar Augusto Fonanillo Lopez & Abdullah Elbi, *On Synthetic Data: A Brief Introduction for Data Protection Law Dummies*, EUR. L. BLOG (Sept. 22, 2022), <https://europeanlawblog.eu/2022/09/22/on-synthetic-data-a-brief-introduction-for-data-protection-law-dummies/>.

²⁰⁵ For instance, the EU’s GDPR, *supra* note 130, defines personal data as “any information relating to an identified or identifiable natural person” (Article 4(1)). Therefore, any information that has been de-identified to a sufficiently robust standard (where reidentification will not be possible using means

claims that synthetic data is anonymous data, and thus is not captured under privacy laws. For instance, a report commissioned by Mostly AI, a synthetic data provider, and published by Harvard Business Review Analytics Services, contains the following:

With... emerging privacy regulations around the world making the sharing of personal information so complicated, if not impossible, synthetic data is vital to support collaboration. As it is fully anonymous, it is exempt from these rules.²⁰⁶

Such categorical claims should be rejected. As with other anonymization techniques, a determination of whether synthetic data infringes data privacy laws requires a context-specific assessment against existing legal standards. Our analysis confirms that whether synthetic data is captured by privacy laws requires us to consider, first, whether it falls within the material scope of a privacy law to begin with, and second, whether it can be brought outside of the scope of this legal framework by being deidentified in a way that satisfies the law's requirements. Data processing is permissible unless specifically regulated.²⁰⁷ Both the sectoral focus of many privacy laws and the way in which they define the term "personally identifiable information" (PII) imply that even their application to collected data may be limited. As we illustrate, the capacity of these laws to capture synthetic data, which is one step further removed from the individual, is more doubtful. To illustrate this, we first provide a brief introduction to the key terms defining the scope of application of data privacy laws, most notably the legal definition of PII and what constitutes deidentified data. We then apply this analysis to two types of synthetic data. The first is a simple example where, in an attempt to anonymize the data, a dataset containing data collected directly from individuals is used to generate a synthetic dataset.²⁰⁸ For our second example, an entirely synthetic dataset (not based on the direct processing of any collected data describing specific individuals) is used to make inferences about an individual. We query in both cases whether the resulting synthetic data falls within the scope of privacy laws.

reasonably likely to be used by any person; recital 26), falls outside its scope. Most other international instruments define personal data in a similar way. *See, e.g.*, The African Union Convention on Cybersecurity and Personal Data Protection, Art. 1; OECD, RECOMMENDATION OF THE COUNCIL CONCERNING GUIDELINES GOVERNING THE PROTECTION OF PRIVACY AND TRANSBORDER FLOWS OF PERSONAL DATA, OECD/LEGAL/0188 (2013), at 1 ("OECD, PRIVACY GUIDELINES")(also requiring that this data pose "a risk to privacy and individual liberties," para 2).

²⁰⁶ Harvard Business Review, *supra* note 30, at 3 (the report does not relate to specific conditions, but includes general statements). Similarly, Nvidia-supported research suggests that as "synthetic data would not be considered identifiable personal data, privacy regulations would not apply, and obligations of additional consent to use the data for secondary purposes would not be required." NVIDIA, *supra* note 37, at 6.

²⁰⁷ As Schwartz and Solove state, "The general approach to information flow in the United States is a "Schillerian" one... "What is not forbidden is allowed." Paul M. Schwartz & Daniel J. Solove, *The PII Problem: Privacy and a New Concept of Personally Identifiable Information*, 86 NYU L. REV. 1814, 1868 (2011).

²⁰⁸ This is sometime referred to as a linear approach. See, for instance, Solow-Niderman who writes that the "linear approach assumes that the individual who cedes control of their data is the same individual potentially affected by the information collection, processing or disclosure." Alicia Solow-Niderman, *Information Privacy and the Inference Economy*, 117 NW. U. L. REV. 357, 404 (2022).

The U.S. does not tie itself to the mast of a single privacy law.²⁰⁹ Rather, privacy regulations comprise a tapestry of legal provisions at the federal and state levels. Such laws have traditionally been mainly concerned with the liberty of citizens vis-à-vis the state, with data processing operations by private entities being given comparatively wide latitude.²¹⁰ Moreover, the First Amendment constrains the development of some privacy laws.²¹¹ Accordingly, personal data processing is largely permissible, subject to compliance with some limited sectoral legal frameworks²¹² and, more recently, state data privacy laws.²¹³ For the moment, the U.S. is “the great international outlier in [Western societies] for data privacy”.²¹⁴

The lack of a single privacy law in the U.S. means that, unlike other significant privacy frameworks like the EU’s GDPR²¹⁵ or the OECD Privacy Guidelines,²¹⁶ there is no unifying concept that defines its applicability. The closest equivalent is PII, a notion that defines the scope and boundaries of numerous federal and state privacy statutes.²¹⁷ Since these laws differ in how they define PII and equivalent terms,²¹⁸ we use a categorization suggested by Schwartz and Solove to structure our analysis. They document three (sometimes inconsistent, sometimes overlapping) ways in which the term PII is defined: tautological (PII is “information which identifies a person”); “non-public” data; and through a bright-line rule which enumerates types of protected data (the “specifics” approach).²¹⁹

Recent developments point to the emergence of a fourth approach²²⁰ to PII.²²¹ The Californian Consumer Privacy Act (CCPA) of 2018 (as amended in 2021) defines

²⁰⁹ Several proposals are currently seriously discussed, most notably the Federal American Data Privacy and Protection Act, H.R. 8152, 117th Congress (2021-2).

²¹⁰ James Q. Whitman, *The Two Western Cultures of Privacy: Dignity Versus Liberty*, 113(6) YALE L. J. 1151 (2004).

²¹¹ See *infra* near fn. 267.

²¹² See examples *infra* in this Section.

²¹³ *Id.*

²¹⁴ Anupam Chander & Paul M. Schwartz, *Privacy and/or Trade*, 90 U. CHI. L. REV. 49 (2023). Interestingly, international firms are starting to voluntarily adopt stricter provisions, in line with the EU’s GDPR, *supra* note 130. See, e.g., Davis & Marrota-Wrugler, *supra* note 174.

²¹⁵ GDPR, *supra* note 130.

²¹⁶ OECD, PRIVACY GUIDELINES, *supra* note 206.

²¹⁷ Such sectoral legislation applies in health care, financial services, certain educational contexts, and credit reporting, among others, and is typically grounded more in consumer protection than in fundamental rights concerns. Anupam Chander, Margot E. Kaminski & William McGeeveran, *Catalyzing Privacy Law*, 105 MINN. L. REV. 1733, 1734 (2021). They note that “[a]s a final backstop, general-purpose consumer protection regulators, such as the Federal Trade Commission (FTC) and state attorneys general, address a subset of cases falling outside any sectoral rules, again largely following a consumer protection model.”

²¹⁸ Schwartz & Solove, *supra* note 207, at 1816, 1827. The emergence of PII as a front-line legal concept and its significance are documented. *Id.*, at 1819-1828.

²¹⁹ Some laws may fall into more than one category.

²²⁰ Schwartz and Solove proposed the adoption of a similar approach (which they termed PII 2.0) which incorporates a category of identifiable information, which is when there is some “non-remote possibility of future identification” based on the data. Schwartz & Solove, *supra* note 207, at 1878. This was further developed in Paul Schwartz & Daniel Solove, *Reconciling Personal Information in the United States and European Union*, 102 CALIF. L. REV. 877 (2014).

²²¹ Not all recently adopted laws represent a rupture with earlier approaches. Indeed, Virginia’s Consumer Data Privacy Act, Va. Code Ann. §§ 59.1-575 (2022), and the Colorado Privacy Act, Colo. Rev. Stat. Ann. § 6-1-1303 (2018), remain firmly in the “publicly available” approach camp. For a comparison of these laws and the CCPA, see Cathy Cosgrove & Sarah Rippey, *Comparison of Comprehensive Data Privacy Laws in Virginia, California and Colorado*, IAPP (July 2021), <https://iapp.org/resources/article/comparison-comprehensive-data-privacy-laws-virginia-california-colorado/>.

personal information in an open-ended way as information that is “capable of being associated with, or could reasonably be linked, directly or indirectly, with a particular consumer or household”.²²² This definition is accompanied by a non-exhaustive list of examples of such personal information. While the CCPA continues to exclude publicly available information from its scope,²²³ it represents a break from the other approaches to PII “by using the real-world potential for identifiability as a touchstone”.²²⁴ The draft proposal of a federal privacy bill, the American Data Privacy and Protection Act, goes along the same path by applying to “information that identifies or is linked or reasonably linkable, alone or in combination with other information, to an individual” or their device.²²⁵ Both laws also incorporate inferences, at least to some extent, within their ambit, as discussed below.²²⁶ We take the CCPA as our exemplar of this new generation of U.S. privacy laws.

Let us now apply these laws to our first case, where synthetic data is generated using collected data about an individual as an input, either in the data generator or as a comparator, to refine the quality of the resulting synthetic dataset. Will the synthetic data produced in such circumstances be captured by privacy laws? This depends on which of the four definitions of PII apply.

At first glance, the CCPA’s definition of PII²²⁷ is most likely to bring such synthetic data within its scope. At the same time, the CCPA excludes deidentified data from its application. Deidentified data is defined as information that cannot reasonably be connected to a particular consumer, provided that business has implemented technical and organizational safeguards to prevent reidentification.²²⁸ In 2023 a new definition came into effect, defining such data as information that “cannot reasonably be used to infer information about, or otherwise be linked to a particular consumer”.²²⁹ Accordingly, what is crucial is whether the synthetic data can reasonably be connected to a specific individual or household. The amendment clarifies that this connection

²²² California Consumer Privacy Act of 2018 (CCPA), Cal. Civil Code §§1798.100-199 (2018).

²²³ Pursuant to the changes introduced by the California Privacy Rights Act of 2020, Cal. Civil Code §§1798.100 (2020), “personal information” does not include publicly available information or lawfully obtained, truthful information that is a matter of public concern. “Publicly available information” is information that is lawfully made available from federal, state, or local government records, or information that a business has a reasonable basis to believe is lawfully made available to the general public by the consumer or from widely distributed media, or by the consumer, or information made available by a person to whom the consumer has disclosed the information if the consumer has not restricted the information to a specific audience.

²²⁴ Chander et al., *supra* note 217, at 1750.

²²⁵ American Data Privacy and Protection Act (ADPPA), H.R. 8152, 117th Congress (2021-2). The CCPA, *supra* note 222, art. 1, defines “covered data” as “information that identifies or is linked or reasonably linkable, alone or in combination with other information to an individual or a device that identifies or is linked or reasonably linkable to an individual and may include derived data and unique identifiers.”

²²⁶ The ADPPA, *Id.*, excludes from the definition of “covered data” inferences made exclusively “from multiple independent sources of publicly available information that do not reveal sensitive covered data with respect to an individual.” This suggests that sensitive inferences based on publicly available information as well as other inferences derived from non-publicly available information remain within the scope of the law. Yet unlike the CCPA, *supra* note 222, it does not capture household data- rather than personal data- within its scope.

²²⁷ CCPA, *supra* note 222, art. 1.

²²⁸ *Id.*, art. 3.

²²⁹ *Id.*, art. 1798.140(m).

exists when one can infer information about a certain consumer or household from the data.²³⁰

Let us delve deeper into the reasonableness requirement. As noted above, the technical literature suggests that where synthetic data is generated using collected data as an input or comparator, then a risk remains that it can be linked back to an individual through inference²³¹ or by linking the synthetic data with other datasets.²³² Yet a question arises as to how much time and effort would be considered reasonable to invest in preventing reidentification. This might require taking into account factors such as who might reasonably be thought to engage in reidentification, for what purpose, and what assumptions should be made with regard to their ability to engage in reidentification. As these factors suggest, what might be deemed a “reasonable” risk threshold might differ from one type of data or one size of dataset to another, and might change over time. Accordingly, how low this threshold should be is a legal and policy question that will have to be determined by courts and legislators. It is noteworthy that in the EU, where a similar test has been in place since 1995,²³³ there remains much confusion and disagreement regarding the acceptable level of risk and how this may be quantified.²³⁴ Note that the level of utility from the dataset and who it benefits does not come into the equation. In addition to this reasonableness requirement, the CCPA now also requires organizational safeguards such as a public commitment not to reverse the deidentification process, unless it is necessary to verify deidentification.²³⁵ It is apparent therefore that synthetic datasets generated using an individual’s collected data can still fall within the scope of the CCPA.

The application of the other statutory definitions of PII to our first type of synthetic data is even more questionable (assuming that the data was not sufficiently deidentified). Take, for example, the definition of “personal information” found in the Children’s Online Privacy Protection Act (COPPA).²³⁶ COPPA defines “personal information” as “individually identifiable information about an individual collected online”. It provides a list of examples, including, among others, an individual’s first and last name, and a home or other physical address. The list of examples – all specific identifiers – suggests that personal information is unlikely to capture our first type of synthetic data.²³⁷ However, the law also enables the FTC to adopt implementing provisions.²³⁸ Furthermore, in 2019 the FTC launched a consultation suggesting that it was willing to consider what is deemed PII, specifically querying whether it should be revised to include information that is “inferred about, but not directly collected from

²³⁰ *Id.*

²³¹ Inference is understood here as the possibility to deduce, with significant probability, the value of an attribute from the values of a set of other attributes.

²³² See discussion *infra* Section III.C.1.

²³³ Identifiability is assessed by reference to the means “reasonably likely” to be used by a controller or another third party to reidentify the individual. GDPR, *supra* note 130, Recital 26.

²³⁴ See, e.g., Michèle Finck & Frank Pallas, *They Who Must Not be Identified—Distinguishing Personal from Non-Personal Data Under the GDPR*, 10(1) INT’L DATA PRIVACY 11 (2020).

²³⁵ CCPA, *supra* note 222. The business must also specifically prohibit reidentification and prevent the inadvertent release of the deidentified dataset.

²³⁶ 15 U.S.C. 91 §6501-6506.

²³⁷ Bellovin et al., *supra* note 108, at 48 (“because privacy statutes do not speak to “fake” data, a door is left open, for better or worse”).

²³⁸ COPPA, *supra* note 236, Rule 90.

children” and “other data that serve as proxies for personal information covered under this definition”²³⁹ Changes along these lines would more readily bring synthetic data generated using collected data within the law’s scope. Yet this consultation closed the same year it opened without any such substantive amendments having been proposed, much less adopted.²⁴⁰ Furthermore, even with such an expanded definition of personal information, COPPA applies only to personal information that is collected online by websites and online services.²⁴¹

Other laws that adopt this “specifics” approach are also unlikely to encompass synthetic data generated using collected data. The Health Insurance Portability and Accountability Act 1996 (HIPAA) adopts an inverse-specifics approach.²⁴² It applies a safe harbor for the sharing of protected health information as long as seventeen specified identifiers are removed from the dataset, including, *inter alia*, names, e-mail addresses, social security numbers, and medical record numbers. Synthetic data is unlikely to contain these real identifiers if the entire dataset has been replaced by artificially generated data. Therefore, even if it might be possible to infer the identity of an outlier individual from the shared dataset, it would be considered sufficiently private for HIPAA. In this sense, the specifics approach could lead to under-inclusive protection.²⁴³ Given that this approach does not allow for a contextual assessment of privacy loss, the protection offered will not be sufficiently calibrated to the risks that our first form of synthetic data might pose.

The tautological approach to defining PII might also not apply to our first case of synthetic data. This can be exemplified by the Video Privacy Protection Act (VPPA), which prohibits “videotape service providers” from knowingly disclosing PII to third parties (with certain exceptions).²⁴⁴ It defines PII as including “information which identifies a person as having requested or obtained specific video materials or services from a video tape service provider.”²⁴⁵ This definition has been interpreted narrowly, to exclude individuals who are identifiable, rather than identified, from its scope. In the *Hulu Privacy Litigation*, the U.S. District Court for the Northern District of California assessed whether Hulu, a provider of online access to pre-recorded content, infringed the VPPA when it provided the URL addresses of content viewed by Hulu users to

²³⁹ FED. TRADE COMM’N, REQUEST FOR PUBLIC COMMENT ON THE FEDERAL TRADE COMMISSION’S IMPLEMENTATION OF THE CHILDREN’S ONLINE PRIVACY PROTECTION RULE, 16 CFR Part 312 RIN 3084–AB20 (2019)(Seeking comments for a new COPPA rule, including a question about whether to include inferred information in the definition of personal information).

²⁴⁰ FTC Commissioner Bedoya suggested that the FTC’s preference is for Congress to amend the law, whether COPPA or by introducing a federal privacy law. Andrea Vittorio, *FTC’s Bedoya Calls for Congress to Update Kids’ Privacy Law*, BLOOMBERG LAW (Sept. 20, 2022), <https://news.bloomberglaw.com/tech-and-telecom-law/ftcs-bedoya-calls-for-congress-to-update-kids-privacy-law?context=article-related>.

²⁴¹ COPPA, *supra* note 236.

²⁴² 42 U.S.C.1320d.

²⁴³ Bellovin et al., *supra* note 108, at 42, argue that it might also be over-inclusive as the risk of linkage across datasets is reduced, as we would be cross-referencing collected data with synthetic data.

²⁴⁴ These exceptions apply when the provider is compelled to provide the information by a warrant; when the provider only discloses names and addresses for the purpose of direct marketing; or when the provider makes the disclosure “incident to the ordinary course of business.” 18 U.S.C. § 2710(b)(2).

²⁴⁵ *Id.*, at § 2710(a)(3). This includes prerecorded video cassette tapes or “similar audio visual materials” and therefore captures the services of content streaming services.

Facebook.²⁴⁶ Despite the obvious risk that Facebook could connect this information with information it already held about its users in order to identify them, the Court held that there was no disclosure of PII as Hulu did not have actual knowledge that Facebook would identify individuals on the basis of the disclosed data.²⁴⁷ In the absence of such actual knowledge, this interpretation enables synthetic data providers to assume that such data would not constitute PII. This assumption could be strengthened by, for instance, introducing contractual restrictions on connecting the synthetic data with individuals in the collected data used to generate it.

The third category of privacy laws, such as the Gramm-Leach-Bliley Act (GLBA), which requires financial institutions to respect the privacy of financial information pertaining to their clients,²⁴⁸ exclude publicly available information from the definition of PII (the “non-public category”).²⁴⁹ The GLBA defines “publicly available” as “any information lawfully available to the general public from federal, state or local governmental records, widely distributed media (including the Internet), or disclosures to the general public required to be made by federal, state or local law”.²⁵⁰ Data about an individual scraped from publicly available websites and subsequently used by financial institutions would therefore not constitute PII.²⁵¹

What emerges from the application of these various laws to our first category of synthetic data is that the newer generation of privacy laws, like the CCPA, are less rigid and more adaptive to technological change than earlier privacy laws. They achieve this, in part, through their broad and general reach, a facet of the law that is not without controversy or complication. In contrast, the more specific sectoral privacy laws apply to this form of synthetic data in much the same way as to collected data. They suffer from under-inclusivity and failure to capture relevant privacy risks, as well as over-inclusivity when they apply irrespective of such risk.

Let us now apply the legal definitions to the second type of synthetic data: replicating collected data by using assumptions rather than the direct processing of collected data. Here the application of privacy laws is even more questionable, despite the fact that the same knowledge about an individual that is otherwise protected by privacy laws could still be derived. Of the laws discussed above, in their current format,

²⁴⁶ *In re Hulu Privacy Litigation*, 86 F. Supp. 3d 1090 (N.D. Cal. 2015).

²⁴⁷ *Id.*

²⁴⁸ 15 U.S.C. § 6809.

²⁴⁹ Charles M. Horn, *Financial Services Privacy at the Start of the 21st Century: A Conceptual Perspective*, 5 89 N.C. BANKING INST. 107, 107 (2001).

²⁵⁰ 16 CFR 314.2(n)(2)(ii)(B). It defines personally identifiable financial information as information a customer provides to obtain a financial product or service from a provider, information about them resulting from such a transaction, or information otherwise obtained about a consumer in connection with the provision of a financial product or service to them. Information that “does not identify a consumer,” including “aggregate information or blind data that does not contain personal identifiers,” is excluded.

²⁵¹ Contrast this with the CCPA, *supra* note 222, where the definition of publicly available information is narrower. There, ‘publicly available’ means information that is lawfully made available from federal, state, or local government records, or information that a business has a reasonable basis to believe is lawfully made available to the general public by the consumer or from widely distributed media; or information made available by a person to whom the consumer has disclosed the information if the consumer has not restricted the information to a specific audience. ‘Publicly available’ does not mean biometric information collected by a business about a consumer without the consumer’s knowledge. CCPA, *supra* note 222, at §1798.140(o)(1)(IK)(2). The application of CCPA to scraping is contested. Brian Stuenkel, *Personal Information and Artificial Intelligence: Website Scraping and the California Consumer Privacy Act*, 19 COLO. TECH. L.J. 429 (2021).

only the CCPA and the American Data Privacy and Protection Act potentially apply to such data.

To show this, it is useful to think of personal data as on a spectrum of proximity to an individual:



Figure 5: *Spectrum of Proximity of Personal Data to an Individual*

Closest to the individual is data that identifies her, followed by data from which she can be identified with additional effort. The next three categories focus on inferences, which we define, following Wachter, as information relating to a natural person “created through deduction or reasoning rather than mere observation or collection from the data subject”.²⁵² These categories relate to the *source* of the inferences. The first includes inference-based synthetic data derived from data pertaining to a specific individual (inferences about Ann based on data describing Ann); the second includes inference-based synthetic data derived from data pertaining to a third party (inferences about Ann based on data describing Barry). The last category relates to inferences about an individual derived from a synthetic dataset pertaining to a group. In the last category it is useful to separate two cases: inferences about Ann based on an effectively deidentified dataset, which was previously based on collected data (pertaining to Ann or others); and inferences about Ann based on a dataset which was generated by a model based on assumptions about how individuals behave, which never related to any specific individual. In all cases the inference is linked back to Ann based on some known fact about her.

The legal treatment of inferences has, until recently, received relatively little attention in privacy scholarship and doctrine.²⁵³ The main question contested was whether an inference deduced about an individual from their own personal data constitutes PII.²⁵⁴ To the extent that the law has been attentive to inferences to date, this has been primarily to consider whether inferences about a person deduced from their own personal data constitute personal data. Most famously, the CCPA gives consumers the right to know specific pieces of personal information that a business has collected about them. The Attorney General (AG) in California was asked to opine on whether this right applied to internally generated inferences from either internal or

²⁵² S. Wachter & Brent Daniel Mittelstadt, *A Right to Reasonable Inferences: Re-Thinking Data Protection Law in the Age of Big Data and AI*, 2019(2) COLUM. BUS. L. REV. 494 ,515 (2019). For an alternative definition of inference see Hallinan & Borgesius, *supra* note 184, at 1 (“Personal data constituting an assertion about a data subject, built on the back of facts about that subject, subjected to some interpretative framework to produce has

²⁵³ The legal classification and governance of inferences is foregrounded in the work of Salóme Viljoen, *A Relational Theory of Data Governance*, 131(2) YALE L. J. 573 (2021) and Solow-Niederman, *supra* note 208. In the EU, the judgment of the CJEU in C-434/16, *Nowak v. Data Protection Commissioner*, EU:C:2017:994, brought inferences within the material scope of EU data protection law, however the consequences of this classification remain disputed. See, e.g., Wachter & Mittelstadt, *supra* note 252.

²⁵⁴ Viljoen, *id.*

external information sources held by the business about the consumer. The AG considered that the CCPA applies to such inferences.²⁵⁵ The definition of “personal information” in the Act includes a subdivision listing the types of information that constitute personal information, such as personal identifiers, consumer information, and online transactions.²⁵⁶ It also includes inferences,²⁵⁷ “drawn from any of the information identified in this subdivision,” used to create consumer profiles.²⁵⁸ The AG reasoned that the source of personal information was irrelevant when responding to a request to know,²⁵⁹ as long as it was of the kind listed in the Act. ²⁶⁰ Accordingly, even inferences based wholly or partly on public records would need to be disclosed, so long as they fall within the list and were used to create a profile of the consumer.²⁶¹ Based on this interpretation of the law, as the list of identifiers is merely an indicative list, it remains possible that inferences about Ann derived from a broad range of information about Ann, constitute her personal information.

It is unclear whether a causal connection is required between the data on Ann and the inference with regard to Ann, and what form this might take. What if, for example, Ann’s data has only a marginal effect, and the same inference could be drawn without Ann’s data, by using Barry’s data? ²⁶² Can the “identified or identifiable” individual be *any* individual who is subject to a data-informed inference? The text of the CCPA leaves this possibility open. In particular, its definition of personal information includes information that *relates to* a particular consumer or household.²⁶³ In the EU, the term “relates to” has been interpreted to mean that the content of the data provides some

²⁵⁵ Opinion No. 20-303 – California Department of Justice, Opinion of Rob Bonta Attorney General and Susan Duncan Lee Deputy Attorney General, 105 Ops. Cal. Atty. Gen. 26, 10 March 2022.

²⁵⁶ The definition of personal information specifically includes personal identifiers (such as name, date of birth and SSN) in addition to information about education, employment, travel, health, credit, banking, Internet Protocol addresses, online transactions, online searches, biometric data, or geolocation data. It also includes ““inferences drawn from any of the information identified in this subdivision to create a profile about a consumer reflecting the consumer’s preferences, characteristics, psychological trends, predispositions, behavior, attitudes, intelligence, abilities, and aptitudes.” CCPA, *supra* note 222, at §1798.140, subd. (o)(1).

²⁵⁷ Inferences are defined as the “derivation of information, data, assumptions, or conclusions from facts, evidence, or another source of information or data.” *Id.*, at §1798.140, subd. (m).

²⁵⁸ A consumer profile is “a profile about a consumer reflecting the consumer’s preferences characteristics, psychological trends, predispositions, behavior, attitudes, intelligence, abilities, and aptitudes.” *Id.*

²⁵⁹ AG Opinion, *supra* note 255, pointing to CCPA, *id.*, subd. (o), which draws no distinction between public and private sources. It follows that, for purposes of responding to a request to know, it does not matter whether the business gathered the information from the consumer, found the information in public repositories, bought the information from a broker, inferred the information through some proprietary process of the business’s own invention, or any combination thereof.

²⁶⁰ The definition of inferences refers to inferences drawn from “information identified in this subdivision”. This suggests that the inference must be based on one of the already specified examples of personal information. One might however argue for a broader construction of this provision: as it refers to the “subdivision” (subd. (o)) and the subdivision includes the definition of personal information. Moreover, the list of personal information examples provided is not exhaustive.

²⁶¹ This rules out situations where a business is using inferences for reasons other than predicting, targeting, or affecting consumer behavior. AG Opinion, *supra* note 255, at 12.

²⁶² As Solow-Niederman explains, machine learning models aggregate the data of individuals to then identify patterns which are subsequently used to make inferences about other individuals who may not have been part of the original dataset. In this way, the models create “inferential pathways...[that]...shift away from “your” data, and towards a new category of “information that might be about you”. Solow-Niederman, *supra* note 208, at 357.

²⁶³ CCPA, *supra* note 222.

information about a person, or that it relates to them in terms of the purposes of its processing or its effects.²⁶⁴ If a similar interpretation were applied under the CCPA, it would capture inferences about an individual derived from the data of others. Indeed, while the inference should be derived from the list of examples of personal information found in the CCPA, the term inference is defined broadly as “the derivation of information, data, assumptions, or conclusions from facts, evidence, or another source of information or data”.²⁶⁵ This suggests that inferences regarding a person generated from entirely artificial data (with no personal data used as a direct input) might still be captured under the law.²⁶⁶

The challenge with this interpretation is, of course, that it may cast the net of privacy laws too wide. Moreover, it could well raise First Amendment challenges. The First Amendment has been interpreted to protect the content of speech between parties, and an inference constitutes such speech. In particular, the speech of private commercial actors – which might include companies making inferences – has historically been treated as protected speech.²⁶⁷ The extent to which states can limit the free speech rights of digital platforms is currently the subject of disagreement between states.²⁶⁸

This analysis demonstrates why any categorical claim that data privacy laws do not apply to synthetic data must be rejected. At the same time, the applicability of rule-based concepts of PII largely fail to incorporate synthetic data processing. Principle-based approaches, like the CCPA and COPPA, offer much more scope for capturing synthetic data, but much depends on how they will be interpreted.

C. ARE DATA PROTECTION LAWS FIT FOR PURPOSE?

The analysis above raises fundamental questions about what we seek to protect through information privacy laws, and whether our methods are fit for purpose. In this section we explore three main normative challenges: the focus on categories of information; the limited ability to capture spillover effects from data on others; and collective data harms. While such challenges are not unique to synthetic data, the rise of synthetic data pushes these tensions and challenges into the spotlight, and may

²⁶⁴ Nowak, *supra* note 253, para. 35.

²⁶⁵ CCPA, *supra* note 222, at § 1798.140 (m).

²⁶⁶ Court of Justice of the European Union, Case C-184/20, Request for a preliminary ruling under Article 267 TFEU from the Vilniaus Apygardos Administracinis Teismas (August 1, 2022)(data that are capable of revealing the sexual orientation of a natural person “by means of an intellectual operation involving comparison or deduction” are in fact sensitive data protected by Article 9 of the GDPR, *supra* note 130. There, the publication of the name of a spouse or partner amounted to the processing of sensitive data because it could reveal sexual orientation, even if no such inference was indeed made).

²⁶⁷ See, e.g., *Reno v. American Civil Liberties Union*, 521 U.S. 844 (1997) or, more recently, *Packingham v. North Carolina*, 582 US __ (2017).

²⁶⁸ Both Florida and Texas sought to introduce laws which prohibited digital platforms from restricting content based on the viewpoint of the user or another person. The Florida law was deemed an unconstitutional interference with the free speech rights of private platforms by the Eleventh Circuit, while the Texas Law was upheld by the Fifth Circuit Court of Appeals. Tom Jowitt, *Florida Seeks Supreme Court Ruling on Social Media Law*, SILICON (Sept. 23, 2022), <https://www.silicon.co.uk/e-management/social-laws/florida-supreme-court-social-media-law-477250>.

exacerbate them. As a result, synthetic data affects the balance between privacy and data utility on which privacy laws are based.

Another way to frame this challenge is through the lens of Nissenbaum's context-relative information norms, which prescribe or proscribe actions relating to the flow of information about an information subject from one actor to another.²⁶⁹ Synthetic data highlights the importance of considering indirect data relations within data flows. In particular, it challenges what Nissenbaum calls "transmission principles," which are constraints on the flow of information from one party to another in a specific context.²⁷⁰ This is because common transmission principles (such as consent and anonymization) might be insufficient to protect privacy in the age of synthetic data. Indeed, if synthetic data can preserve the utility of a dataset while dispensing with the need for direct use of collected data to generate it, this may serve to circumvent rather than promote the objectives of existing privacy rules.

1. *Challenges arising from categorizing data*

Privacy laws often protect categories of information-sensitive data, non-public data, or specific types of data.²⁷¹ As Ohm has written, this approach "assumes that lawmakers can evaluate the inherent riskiness of data categories, assessing with mathematical precision whether or not a particular data field contributes to the problem enough to be regulated".²⁷² Yet, as our discussion indicates, this approach fails to capture all types of data that might create privacy harms.

Furthermore, this approach gives insufficient weight to the fact that deidentified data – such as synthetic data generated either by using collected data directly or by using inferences based on real-world observations – might be linked back to individuals.²⁷³ The unreliable boundary between deidentified and identified data is widely recognized in the legal and computer science literatures. For example, already in 2009, Acquisti and Gross showed how it was possible to predict an individual's social security number from only publicly available data about her place and date of birth.²⁷⁴ As the accumulated knowledge mined for data grows, and better data-mining techniques are created, this reverse-engineering problem will only increase.²⁷⁵ Accordingly, the risk of

²⁶⁹ HELEN NISSENBAUM, *PRIVACY IN CONTEXT: TECHNOLOGY, POLICY, AND THE INTEGRITY OF SOCIAL LIFE* (2009).

²⁷⁰ *Id.*

²⁷¹ For example, the ADPPA, *supra* note 225, refers to "covered sensitive data".

²⁷² Ohm, *Broken Promises*, *supra* note 180, at 1734. Zarsky notes that "the rise of big data substantially undermines the logic and utility of applying a separate and expansive legal regime to special categories." Tal Zarsky, *Incompatible: The GDPR in the Age of Big Data* (2017) 47 SETON HALL L. REV. 1014. De Gregorio notes that the "rationale behind the distinction between 'ordinary' and 'particular' categories of data tends to be nullified by the way in which the data are processed." GIOVANNI DE GREGORIO, *DIGITAL CONSTITUTIONALISM IN EUROPE* 243 (2022).

²⁷³ Given the potentials of big data analytics, which allows to draw probability conclusions from correlations between different pieces of information, it is no longer possible to neatly distinguish between non-personal and personal data. JOSEF DREXL, *DATA ACCESS AND CONTROL IN THE ERA OF CONNECTED DEVICES* 124 (2019).

²⁷⁴ "The inferences are made possible by the public availability of the Social Security Administration's Death Master File and the widespread accessibility of personal information from multiple sources, such as data brokers or profiles on social networking sites." Alessandro Acquisti & Ralph Gross, *Predicting Social Security Numbers from Public Data*, 106(27) PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES 10975 (2009).

²⁷⁵ As Schwartz and Solove state, the "public or private status of data often does not match up to whether it can identify a person or not," and individuals may not want their publicly available data aggregated with

reidentification will grow along with the availability of more collected data, and of synthetic datasets based on such data. Cloud storage, which significantly reduced the costs of data storage, further magnifies this possibility.

In light of such limitations of a category-based approach to information privacy, one might query whether an approach that focuses on the final, synthetic dataset rather than the category of data it includes (such as an individual's social security number) might be superior. This proposition faces two main obstacles. From a practical perspective, this would require an overhaul of some existing statutory instruments, changing both the designated entity (such as videotape service providers in the VPPA or financial institutions in the GLBA) and the focus of the limitations on PII. Second, the risk of reidentifiability may vary with conditions that the generator of the dataset might not be privy to (for example, how much more collected data the user has or is likely to have). This risk could also change over time. In line with this challenge, the UK data protection regulator argued that it is almost impossible to predict the risk of reidentification through data linkage, as "it can never be assessed with certainty what data is already available or what data may be released in the future".²⁷⁶ Accordingly, imposing such a requirement might significantly harm the ability to use any synthetic data relating to people, unless applied in a manner which is sensitive to the effects of such changes on incentives for beneficial data mining.

Another suggestion involves imposing statutory obligations on the downstream operators (such as Facebook in the Hulu litigation) which can turn identifiable data into identified information. The fact that the probability of identifiability may change over time strengthens this proposition. Indeed, to limit over-broad chilling effects on the use of data, such limitations should potentially balance the probability of detection at the time the dataset was internally created or purchased, the costs and financial risks involved in creating it, the observable level of harm at the time of use, and the social utility from that can be gained by data flows via synthetic datasets.

2. Limited ability to capture spillover effects

The next two challenges raise more fundamental conceptual questions regarding the connection between the data governed by data privacy laws and the objectives of these laws. Most obviously, are we concerned with protecting the data of an individual as such (such as the health information of Ann), or are we concerned with the impact that data processing might have on power dynamics between the data holder and the individual? As our analysis demonstrates, data privacy laws tend to emphasize the personal nature of the information processed – focusing primarily on the *source* or *type* of data at stake – rather than on the nature of the harms that might flow from data processing. They are predicated on an implicit assumption that by protecting PII, we are protecting the privacy rights of each individual. More importantly, for harm to arise, there must be a direct connection between the individual and the personal information protected, in the

non-publicly available information. Schwartz & Solove, *supra*, note 207, at 1830. See further, Helen Nissenbaum, *Protecting Privacy in an Information Age: The Problem of Privacy in Public*, 17 L. AND PHIL. 559 (1998).

²⁷⁶ INFORMATION COMMISSIONER'S OFFICE, ANONYMISATION: MANAGING DATA PROTECTION RISK, CODE OF PRACTICE 18 (2012).

form of direct or indirect identifiers or inferences. Synthetic data challenges this assumption, requiring us to re-assess the nature of the connection between the data processed and the individual. Synthetic data can increase such concerns in two main ways: through the ability to learn via inferences based on data of others (spillover effects), and by strengthening collective data harms.²⁷⁷

Synthetic data increases spillover data privacy harms, which are beginning to be recognized, and which further challenge existing assumptions regarding the source of data.²⁷⁸ Such harms result from externalities in data analysis. For example, an individual's data (provided or observed) may enable the accumulation of data about others as well. Put differently, maintaining the confidentiality of Ann's data will not necessarily prevent those details from being inferred from (public) data on Barry. For instance, if we know from data on others that eating a high-fat diet increases the risk of heart conditions, and that Ann is part of a community that eats a high-fat diet, we may infer that Ann is at higher risk of heart disease. As Viljoen argues, this relationality, or ability to make inferences about others from data, is not simply a negative externality of current business models; rather, it is integral to them and "constitutes much of what makes data production economically valuable in the first place."²⁷⁹ Solow-Niederman takes this point one step further: "Machine learning... facilitates an inference economy that strains this protective approach past its breaking point. Machine learning provides pathways to use data and make probabilistic predictions – inferences – that are inadequately addressed by the current regime."²⁸⁰ Accordingly, as Tontrup and Sprigman argue, data analysis externalities "strip the individual of the power to protect her privacy alone".²⁸¹

Synthetic data increases these risks. Take, for example, collaborative filtering, which is based on correlations in data groups (for example, consumers and products).²⁸² Data on previous purchases and the features of consumers creates data spillovers among data subjects, by enabling the algorithm to make inferences about Ann based on collected data relating to Barry. Furthermore, even if a synthetic dataset anonymizes information about each individual but enables the algorithm to learn about groups, once the algorithm can connect an individual to a group, it can make informed inferences about her preferences. This begs the question of whether the concept of identifiability is sufficient to prevent harm to individuals, and whether it can capture linkages or inferences on which synthetic data might be based. This also casts further doubt on the utility of individual control over one's privacy. Accordingly, lawmakers and courts face a dilemma: to define or interpret the scope of data privacy laws more broadly, thereby loosening the link between collected data and the individual and capturing more data

²⁷⁷ On the distinction between group privacy and collective privacy see Alessandro Mantelero, *From Group Privacy to Collective Privacy: Towards a New Dimension of Privacy and Data Protection in the Big Data Era*, in GROUP PRIVACY (Linnet et al. eds., 2017).

²⁷⁸ See, e.g., Kate Crawford & Jason Schultz, *Big Data and Due Process: Toward a Framework to Redress Predictive Privacy Harms*, 55 B.C. L. REV. 93 (2014); DREXL, *supra* note 273, at 124 ("Given the potentials of big data analytics, which allows to draw probability conclusions from correlations between different pieces of information, it is no longer possible to neatly distinguish between non-personal and personal data"); Madison et al., *supra* note 134 **Error! Bookmark not defined.**, at 7.

²⁷⁹ Viljoen, *supra* note 253, at 611.

²⁸⁰ Solow-Niederman, *supra* note 208, at 357.

²⁸¹ Tontrup and Sprigman, *supra* note 203, at 2.

²⁸² See *supra* Section II.

flows under their scope, or to see some of the values that data privacy laws promote undercut by synthetic data processing.

3. *Collective data harms*

For similar reasons, synthetic data also increases collective data harms. Such harms arise when the analysis of data leads to decisions that might affect a group of individuals (such as residents of a town or a country), whose data may or may not constitute part of the dataset.²⁸³ A well-known example involves the Facebook/Cambridge Analytics debacle,²⁸⁴ where granting access to data on one individual led to revealing collected data related to others. The data accumulated might have potentially led to manipulations that affected the political system, thereby indirectly impacting a group of individuals (all U.S. citizens),²⁸⁵ with no need to include or tie each individual in this group to the dataset. The rise of synthetic data highlights the fact that even if anonymization of a dataset can be increased, privacy harms might also increase, given that now more non-personal data may be available for use.

Accordingly, synthetic data challenges the effectiveness of existing data privacy laws in significant ways, calling into question their rationale: whether their goal is to protect the data of an individual²⁸⁶ or to protect society from (certain) information-induced harms.²⁸⁷ It also challenges the efficiency, and thus the continued relevance, of current data privacy laws in achieving an overall positive social-welfare balance, once the effects of synthetic data are added to the equation.

The need to reexamine current data privacy laws is heightened by the fact that findings with regard to the (in)applicability of privacy laws to synthetic datasets could affect informed data subjects' incentives to share or protect their data. Synthetic data could increase rational apathy towards data protection: if the law does not protect against the use of synthetic data in ways that might negatively affect citizens or consumers, and if data subjects have only a marginal ability to affect the collection of data that could serve as the basis for synthetic data generation, then individuals have no incentive to invest in data protection. This could increase individual and collective harms.

Indeed, if we assume that a core concern of privacy law is to act as a constraint on informational power – the power that an entity derives from having significant knowledge about an individual or group – then it matters little whether the source of the information is a direct identifier or a proxy for it. This leads us to ask, what should be the appropriate relational link with an individual for information to fall within the

²⁸³ See generally Solow-Niederman, *supra* note 208.

²⁸⁴ Alvin Chang, *The Facebook and Cambridge Analytica Scandal, Explained with a Simple Diagram*, VOX (May 2, 2018), <https://www.vox.com/policy-and-politics/2018/3/23/17151916/facebook-cambridge-analytica-trump-diagram>.

²⁸⁵ BRITTANY KAISER, *TARGETED: THE CAMBRIDGE ANALYTICA WHISTLEBLOWER'S INSIDE STORY OF HOW BIG DATA, TRUMP AND FACEBOOK BROKE DEMOCRACY AND HOW IT CAN HAPPEN AGAIN* (2019).

²⁸⁶ This conceptualization of “data as an individual medium” privileges data processing's capacity to transmit knowledge about the data subject over its capacity to transmit knowledge about others. Viljoen, *supra* note 253, at 594.

²⁸⁷ Among the values we might protect through data protection are autonomy, dignity, identity, freedom of conduct, and democracy.

protective ambit of our laws? If we would like to protect individuals' privacy, these harms need to be addressed. Indeed, if the synthetic generation process is successful, then the dataset generated will constitute a convincing replica of a dataset about real-world people. If this replica dataset can be used to impact individuals, then irrespective of the precise data used to draw this inference, the threat to individuals' fundamental rights will be the same. Put differently, "it does not matter *who* the data 'came' from, but *what* such data says about [a person], and *how* such meaning is used to act upon [a person]".²⁸⁸ The European Court of Justice, reasoning along these lines, adopted a purposive interpretation of its privacy regulation, the GDPR. By this logic, "the boundaries of the concept of information in data protection eventually stretch to encompass whatever substance engages the types of harms dealt with by data protection".²⁸⁹ The Court could therefore find that when an inference is brought to bear on an individual, it does not matter whether that inference was derived from other personal data, the data of a third party, or synthetic data.²⁹⁰

Broadening privacy laws to capture these types of datasets might have intuitive appeal. Yet whether the net of privacy law should be cast so widely remains highly contested.²⁹¹ Capturing entirely artificial data under data protection rules would potentially give individuals rights, such as rights to access and delete, in relation to such data – a possibility far removed from the image of a personal information dossier that underlies the idea of individuals' rights over their data. Furthermore, casting the net too widely would significantly restrain the utility of data. Finding an optimal balance is further complicated by the fact that inferences might involve minimal privacy loss or harm in some contexts (e.g., if Ann's colleagues use Ann's lunch choices to make inferences about her diet, and thus her health), but very significant harm in others (if an insurance company uses the contents of Ann's shopping trolley to do the same).²⁹²

The complex relationship between personal data protection and goals like guaranteeing functioning markets and enhancing innovation has long been recognized.²⁹³ On the one hand, as Drexel notes, privacy protection can be regarded a condition for the functioning of markets, as well as a driver of innovation. This is because consumers will be less willing to provide their data, and even to buy goods which they assume will affect their personal profile, if the law does not guarantee certain levels of data protection.²⁹⁴ On the other hand, restrictions on the use of data may limit firms' ability to develop innovative products and processes. This has led to a balance whereby certain collection and use restrictions are imposed on personally identifiable

²⁸⁸ Stephen Saxby, *Data Protection Act 1984: Security and Accuracy of Data*, 2(2) COMPUTER LAW AND SECURITY REPORT 14 (1986).

²⁸⁹ Dara Hallinan, *Data Protection without Data: Could Data Protection Law Apply without Personal Data Being Processed?*, 5 EUR. DATA PROTECTION L. REV. 293, 297 (2019).

²⁹⁰ For an excellent overview of the developments leading to international data privacy norms see GLORIA GONZALEZ FUSTER, *THE EMERGENCE OF PERSONAL DATA PROTECTION AS A FUNDAMENTAL RIGHT OF THE EU* 75-107 (2014).

²⁹¹ We reserve this exploration of the role of inference in data protection law for future work.

²⁹² See generally NISSENBAUM, *supra* note 269.

²⁹³ DREXL, *supra* note 273, at 7.

²⁹⁴ *Id.* See also Niva Elkin-Koren & Michal S. Gal, *The Chilling Effects of Governance-by-Data on Data Markets*, 86 U. CHI. L. REV. 403 (2019).

information.²⁹⁵ But this balance is uprooted once synthetic data enters the equation, as it potentially changes both sides of the scale: it limits protections for individuals, and it has the potential to increase competition and innovation.

Until we find the correct balance in the scope of our privacy laws, a task which is beyond the scope of this article, other laws may address some of these concerns indirectly. We turn to some relevant examples in the next section.

V. EFFECTS OF INCREASED DATA QUALITY

Synthetic data can potentially increase the quality of some datasets. Here we briefly explore how synthetic data does this, as well as the effects of such increased quality on data power and social welfare. In particular, we focus on its capacity to alter the relationship between data-based decision-makers and those affected by such decisions, in unique and fundamental ways. We then analyze the ways in which the law affects the motivation to create more accurate and complete datasets, as well as the ability to use them. As we show, while the law has long been attentive to the fact that higher quality data can increase the accuracy of decision-making, not enough consideration has been given to higher quality data as a source of power.²⁹⁶ In particular, the law currently places only limited restrictions on the ability to exploit or manipulate high-quality data in ways which negatively affect individuals or groups.²⁹⁷ As such, synthetic data strengthens the need to consider the effects of increased data quality on data governance. While our analysis applies to both personal and non-personal synthetic data, and to its use in the private and the public spheres, it focuses mainly on the effects on individuals.

A. THE EFFECTS OF SYNTHETIC DATA ON DATA QUALITY

Data quality has multiple related yet distinct dimensions, of which two fundamental elements are completeness and accuracy. Completeness ensures that certain data features are not *unrepresented* in a dataset, while accuracy ensures that they are not *misrepresented* in the dataset.²⁹⁸

Synthetic data offers the potential to strengthen both dimensions.²⁹⁹ It does so by plugging gaps in datasets that result from difficulties in gaining access to collected data,

²⁹⁵ For example, the CCPA contains a “do not sell” rule which enables Californian residents to opt-out of the “selling, renting, releasing, disclosing, disseminating, making available, transferring or otherwise communicating” of their data; and a right to opt out. CCPA, *supra* note 222, at 1798.140(ad)(1) and 1798.120(d).

²⁹⁶ See also Nielsen, *supra* note 25.

²⁹⁷ See *infra* Section IV.B.

²⁹⁸ Accuracy is defined differently in different fields. For our purposes, we use the definition used in machine learning, which is based on the fraction of outputs of a model that are correct. Such a definition, however, is blind to the different (social) weights of different data points. Furthermore, it might mask poor decision-making, if the given dataset is unbalanced. Aileen Nielsen, *Accuracy Bounding: A Regulatory Solution for the Algorithmic Society* (on file with author, 2022), at 7-9. For that, upsampling might be useful.

²⁹⁹ Data may never reach a complete level of accuracy as a picture of the real world, given data collection or generation limitations. Yet it might be accurate with regard to the specific features it includes.

which might emerge when collected data is rare, or too costly or impractical to collect. Take, for example, completeness. During the Covid-19 pandemic, social distancing restrictions placed limits on data gathering for autonomous vehicle training. Google's Waymo responded by using synthetic data simulations of road conditions, based on data already collected, to continue training during this period.³⁰⁰ Another example centers on efforts for de-biasing machine learning algorithms.³⁰¹ Crawford describes "dark zones or shadows" in datasets where "some citizens and communities are overlooked or underrepresented".³⁰² By augmenting collected datasets in a targeted way, synthetic data can be used to make datasets more representative, reducing the harms stemming from sampling or statistical bias.³⁰³ The potential to synthesize less-biased data to train unbiased or minimally biased models in a variety of contexts, ranging from uneven gender and age distribution to regional biases, is emphasized in the literature on synthetic data.³⁰⁴ The example of Amazon's algorithm for vetting job applicants is a case in point.³⁰⁵ Such bias, which stems from training an automated system on a biased or incomplete dataset, can be addressed by adding synthetic data so that the dataset more accurately reflects either the real world, or the actual parameters that satisfy the needs of the decision-maker (e.g., choosing the best workers). In the same way, synthetic data can also be used to potentially correct over-representation (e.g., the over-policing of certain communities, leading to their over-representation in criminal conduct datasets).

Now consider accuracy. As elaborated in Section II above, by acting as a (partial) replacement for missing data, synthetic data can potentially create more representative datasets. Furthermore, synthetic data can increase accuracy by verifying the correctness of the analysis performed on collected data, as exemplified by the use of synthetic data to create counterfactuals to fix over-confident AI models.³⁰⁶

Accordingly, synthetic data is a potentially useful technological tool to increase data quality. Yet in doing so, synthetic data draws to the fore a fundamental tension.³⁰⁷ On the one hand, as noted, it may increase the quality of decisions and reduce bias based on misrepresentation. Furthermore, by potentially enabling more players to enter the market, it can indirectly increase quality by strengthening market-based motivations to

³⁰⁰ Kyle Wiggers, *Challenges of Developing Autonomous Vehicles During Coronavirus Covid19 Pandemic*, VENTURE BEAT (April 28, 2020), <https://venturebeat.com/2020/04/28/challenges-of-developing-autonomous-vehicles-during-coronavirus-covid-19-pandemic/>.

³⁰¹ There is an extensive literature on bias in automated decision making. See, e.g., VIRGINIA EUBANKS, *AUTOMATING INEQUALITY: HOW HIGH-TECH TOOLS PROFILE, POLICE, AND PUNISH THE POOR* (2018); SAFIYA UMOJA NOBLE, *ALGORITHMS OF OPPRESSION: HOW SEARCH ENGINES REINFORCE RACISM* (2018); Barocas & Selbst, *supra* note 106, at 677-93.

³⁰² Kate Crawford, *Think Again: Big Data*, FOREIGN POLICY (May 10, 2013), <https://foreignpolicy.com/2013/05/10/think-again-big-data/>.

³⁰³ Realistic synthetic data, along with appropriate data imputation techniques offer a promising approach to tackle this challenge. Aseefa et al., *Generating Synthetic Data in Finance: Opportunities, Challenges and Pitfalls*, in ICAIF 20L PROCEEDINGS OF THE FIRST ACM INTERNATIONAL CONFERENCE ON AI IN FINANCE 1, 2 (2020).

³⁰⁴ Harvard Business Review, *supra* note 30, at 6 ("It's not only bias in people—gender, race, and so on—but business biases, such as 'we will pay more attention to this region versus that region because we have more records from this region.'").

³⁰⁵ See *supra* Section II.B.2.

³⁰⁶ Singla et al., *supra* note 85.

³⁰⁷ For such tension with regard to collected data see, e.g., Nielsen, *supra* note 25.

provide higher-quality products and services.³⁰⁸ On the other hand, a high-quality dataset could become a double-edged sword, as more accurate decisions might not always increase social welfare.³⁰⁹ This is a non-trivial claim.³¹⁰ As Chen notes, overly accurate information can enable new forms of differentiation and categorization, which might have negative welfare effects on individuals and groups through exploitation or manipulation.³¹¹ Accurate data can also give rise to a “loss of maneuver space” for individuals.³¹² Likewise, it makes individuals and society more “readable”, potentially reducing individuals’ capacity for self-development and change, while exacerbating power and information asymmetries between those who process data and those who are subject to this data processing. Nielsen’s thoughtful taxonomy of algorithmic accuracy harms is also relevant to more accurate data. She relates to three categories: accuracy directly creating harms (such as undermining human autonomy); behavior associated with the pursuit of accuracy causing harms (side effects, such as privacy incursions in the acquisition and use of data); and strategic responses to algorithms driven by potentially mistaken perceptions (such as automation bias based on assumptions of algorithmic superiority).³¹³ More accurate data can increase all three. Such dangers are, of course, not unique to synthetic data. However, synthetic data exacerbates the regulatory challenge, bringing it to another level which might require a new balance between the competing considerations.

The potential harms of higher-quality data are best illustrated by synthetic data’s potential contribution to the creation of more accurate digital profiles, which, in turn, enable more personalized treatment of individuals.³¹⁴ In the economic sphere, an individual may receive micro-targeted offers for products that better fit her preferences, but possibly at higher, discriminatory prices that reflect her elasticity of demand.³¹⁵ In the social sphere, she may receive suggestions for connections (e.g. via LinkedIn) and content that better cater to her prior interests, but could potentially limit her viewpoints.³¹⁶ In the political sphere, her personalized digital feed could be designed to

³⁰⁸ Shoshana Zuboff, *Big Other: Surveillance Capitalism and the Prospects of an Information Civilization*, 30 J. INFO. TECH. 75 (2015).

³⁰⁹ Furthermore, as Nielsen argues, the current overemphasis on algorithmic accuracy is itself a form of welfare loss, in which practitioners put their efforts exclusively on a proxy that could at times be anticorrelated with social welfare. Nielsen, *supra* note 298.

³¹⁰ The limited legal academic treatment of the principle of data quality so far has mainly focused on accuracy – that data should be correct and precise. See sources cited in this section. However, data may be accurate but still incomplete. Paul Ohm & Jonathan Frankle, *Desirable Inefficiency*, 70 FLA. L. REV. 777 (2018).

³¹¹ Jiahong Chen, *The Dangers of Accuracy*, 36 EUR. DATA PROTECTION L. REV. 42 (2018).

³¹² *Id.*

³¹³ Nielsen, *supra* note 25, at 64.

³¹⁴ See EXEC. OFFICE OF THE PRESIDENT, BIG DATA: SEIZING OPPORTUNITIES, PRESERVING VALUES 7, 44 (2014) (outlining the type of data contained within a profile and the method by which profiles are created); see generally EXEC. OFFICE OF THE PRESIDENT, BIG DATA: A REPORT ON ALGORITHMIC SYSTEMS, OPPORTUNITY, AND CIVIL RIGHTS (2016) (addressing harms that can result from supplying data to algorithmic profiling software); Nielsen, *supra* note 25.

³¹⁵ See FED. TRADE COMM’N, BIG DATA: A TOOL FOR INCLUSION OR EXCLUSION? 9–11 (2016) (outlining ways in which the use of big data can generate harmful consequences for low-income groups).

³¹⁶ See Christoph B. Graber, *The Future of Online Content Personalisation: Technology, Law and Digital Freedoms* 6–8 (Univ. of Zurich, I-Call Working Paper No. 2016/01, 2016) (discussing online content personalization and popular criticisms of it).

strengthen certain opinions and affect her political choices.³¹⁷ In the legal sphere, digital profiles could inform decisions made by law enforcement or judicial bodies (e.g., based on a suspect's presumed flight risk), and even lead to the creation of personalized laws.³¹⁸ While the ability to create a personal profile from a mosaic of data points has long been acknowledged,³¹⁹ synthetic data can increase it dramatically by overcoming barriers in the data value chain. These examples also illustrate that the same data can be used in both welfare-enhancing and welfare-reducing ways.³²⁰

Synthetic data also raises the opposite concern: it might reduce data quality when an analyst bases the generated synthetic data on incorrect assumptions. While many such instances can be addressed by technological means – including by testing the data against collected data or against counterfactuals –³²¹ in others the concern that the quality of data used for decision-making might be reduced, is a real one.

B. APPLICATION OF LAWS

The potential effects of synthetic data on data quality require us to determine to what extent current data governance laws that relate to data quality apply, and whether such application furthers our social goals. Optimally, the law should encourage those instances in which more accurate data-based decisions increase welfare, while prohibiting those in which it significantly reduces it. This, in turn, requires the design of rules that can separate the two types in a cost-effective way – a tall order. Nonetheless, at least in some instances this may be achievable. As we show below, while some laws are fit for purpose, others might need to be changed in order to incorporate this new data generation reality.

We identify five challenges. In the first, laws relating to data quality that are conditioned on the provenance and nature of data – such as data privacy laws – might not apply to synthetic data, despite the fact that their application may have increased social welfare.³²² This leads to the second challenge: Where the provenance or nature of the data, including its method of collection or generation, is the distinguishing parameter for the application of a law, synthetic data increases enforcement challenges. This is because if both real and synthetic data can lead to similar decisions, the enforcer might not be able to distinguish which type of data (such as private data) was used, based on the observed outcome alone. Furthermore, firms might falsely claim to be using a synthetic data generator – a claim which might be hard to contradict without sophisticated reverse engineering.

The third challenge relates to whether and when the law should incentivize or mandate the use of synthetic data through legal requirements (such as **reasonableness**

³¹⁷ See, e.g., Emma Graham-Harrison, Carole Cadwalladr & Hilary Osborne, *Cambridge Analytica Boasts of Dirty Tricks to Swing Elections*, GUARDIAN (Mar. 19, 2018), <https://www.theguardian.com/uk-news/2018/mar/19/cambridge-analytica-execs-boast-dirty-tricks-honey-traps-elections>.

³¹⁸ See generally Omri Ben-Shahar & Ariel Porat, *Personalizing Negligence Law*, 91 NYU. L. REV. 627 (2016) (arguing that courts can and should use data to create personalized reasonable person standards for negligence inquiries).

³¹⁹ Yuichiro Tsuji, *Medical Big Data in Japan*, 8 J.L. & CYBER WARFARE 153, 154 (2020).

³²⁰ See also Nielsen, *supra* note 25.

³²¹ See *supra* Section II.

³²² See *supra* Section IV.

or **risk reduction** requirements), where such data can further legal goals or requirements. As noted, synthetic data widens the scope of possible options: it is no longer necessary to choose whether or not reliance on the collected data accumulated is sufficient to meet the legal requirements. Whether and when these additional options should be taken into account in a legal analysis depends, in part, on synthetic data's technical capabilities. Most evidently, synthetic data should not be treated as a quick or even the most efficient fix for all illegal data-based decisions. Take, for example, illegal discrimination. The introduction of bias through the data sample used to train a model is just one way to create bias. It might equally be introduced, for example, at the point at which the target variable – the objective of the data mining, such as finding a creditworthy borrower – is defined, or when the characteristics associated with that variable (the class labels) are chosen. Another issue involves measurement limitations. To illustrate, the current state of the art of “fairness” in automated decision-making often equates it to parity between two groups. This implies that complex issues such as intersectional discrimination are treated in a simplistic manner that flattens the interests at stake.³²³ More fundamentally, treating synthetic data as a means to tackle discrimination maintains a legal and policy focus on technical fixes.³²⁴ In so doing, we delegate to technologists the task of determining what counts as discrimination and what constitutes a representative dataset. Furthermore, we view the question of bias through a narrow lens.³²⁵ As Chander et al. suggest, “framing the debate around technical responses will obscure the complexity of the impact of AI systems in a broader political economy and ringfence the potential responses to the technical sphere”.³²⁶ In short, using synthetic data to correct a biased dataset is better conceived as a minimum patch to address a flaw, rather than a holistic response to a complex legal and societal problem. Yet, where synthetic data can be relatively easily and cost effectively used to reduce illegal harms, it should be taken into account by courts when considering the reasonableness of the conduct (such as reducing harms or self-help in tort law), or meeting quality-related requirements. Furthermore, it should also affect data governance requirements where accuracy is an important parameter, such as in content moderation practices.³²⁷ In such instances, using synthetic data to increase quality, where such use is possible, is not only responsible, but should also be mandatory.

³²³ AGATHE BALAYN & SEDA GÜRSSES, IF AI IS THE PROBLEM, IS DEBIASING THE SOLUTION? 19 (2021).

³²⁴ Julia Powles, *The Seductive Diversion of “Solving” Bias in Artificial Intelligence*, MEDIUM (Dec. 7, 2018), <https://onezero.medium.com/the-seductive-diversion-of-solving-bias-in-artificial-intelligence-890df5e5ef53>.

³²⁵ BALAYN & GÜRSSES, *supra* note 323, at 121.

³²⁶ Chander et al., *supra* note 217.

³²⁷ Danielle K. Citron & Benjamin Wittes, *The Internet Will Not Break: Denying Bad Samaritans § 230 Immunity*, 86 FORD. L. REV. 401 (2017); Danielle K. Citron, *How To Fix Section 230*, *forthcoming*, BOS. U. L. REV. (2023).

The fourth challenge focuses on legal requirements of explainability and interpretability,³²⁸ designed to further legal norms of transparency and reason-giving,³²⁹ which, in turn, increase accountability. Generally, the more sophisticated the synthetic data generator, the more difficult it becomes to explain correlations, and – even more strongly – causality, in the data generated.³³⁰ Some correct explanations might even be nonintuitive when compared to common-sense understandings of how the world works.³³¹ As a result, synthetic data could strengthen the transparency deficit, reducing the ability of third parties to separate high-quality data from entropy.

Where explainability is a mandatory legal requirement, the use of synthetic data might be limited, reducing the ability to enjoy its benefits. To illustrate, empirical research suggests that when synthetic data is combined with differential privacy, it may offer better privacy protection than traditional sanitization methods for some datasets.³³² However, it is not possible to predict which patterns in the dataset will be preserved, which could also lead to poor interpretability.³³³

To ensure accountability while enabling socially beneficial uses of synthetic data, we propose, instead, that for most uses of synthetic data for which explainability or interpretability are highly costly or impractical, accountability should instead relate to the data generation process, including a quality assurance component. Put differently, accountability and transparency should build more on the “ingredients and machinations” of the processes leading to the data-based decision, rather than the exact internal processes. Accordingly, accountability should relate to the choice of tool that created the synthetic data, the data inputs and assumptions used in the process, and the coder’s modelling choices. It should thus be determined by technically-informed experts; and it should not be assessed in the abstract, but rather should relate to the specific use of the data. Similar proposals were made in the context of AI.³³⁴ This may lead to what Sanfilippo, Strandburg and Frischman call **procedural legitimacy**.³³⁵ Such a focus will also reduce exposure of trade secrets or privacy concerns.

To achieve procedural legitimacy, additional tools might need to be developed. These include standards for synthetic data generation, strengthening accuracy-by-design. Naturally, creating such standards carries some costs. While some elements of data generation standards will be relevant for many contexts, others might differ from one context to another, due to differences in the necessary data and levels of risk. Other

³²⁸ Russell and Norvig consider a machine learning model to be interpretable if “you can inspect the actual model and understand why it got a particular answer for a given input, and how you would change the answer when the input changes.” RUSSELL & NORVIG, *supra* note 196, at 729. For a different definition see, e.g., Cynthia Rudin, *Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead*, 1 NATURE MACHINE INTELLIGENCE 206 (2019).

³²⁹ For some of the justifications for such requirement see, e.g., Jonathan Zittrain, *Intellectual Debt: With Great Power Comes Great Ignorance*, MEDIUM (July 24, 2019), <https://medium.com/berkman-klein-center/from-technical-debt-to-intellectual-debt-in-ai-e05ac56a502c>.

³³⁰ See, e.g., DAVID FREEMAN ENGSTROM, DANIEL E. HO, CATHERINE M. SHARKEY, & MARIANO-FLORENTINO CUÉLLAR, GOVERNMENT BY ALGORITHM: ARTIFICIAL INTELLIGENCE IN FEDERAL ADMINISTRATIVE AGENCIES 75 (2020).

³³¹ *Id.*

³³² Stadler et al, *supra* note 112, at 1.

³³³ *Id.*

³³⁴ Rita Matulionyte, Paul Nolan, Farah Magrabi & Amin Beheshti, *Should AI-Enabled Medical Devices be Explainable?* (Working paper, 2022), <https://ssrn.com/abstract=4140234>.

³³⁵ Madison, Frischmann, Sanfilippo & Strandburg, *supra* note 134, at 7.

costs may arise from the standardization process itself, such as setting sub-optimal standards and capture by strong players.³³⁶ Another tool, suggested by Freeman, Ho, Sharkey and Cuéllar in the context of AI, requires users to engage in prospective “benchmarking” of full or partial datasets “by reserving a random hold-out sample of cases for human decision, thus providing critical information to smoke out when an algorithm has gone astray”.³³⁷ The use of contrarian algorithms, which test the robustness of the explanation of the generation method, may serve similar purposes. At the same time, procedural legitimacy might not be fit for all contexts in which synthetic data can be used. For instance, we might insist that where explainability is essential, synthetic data should only be used as training data but not as test data.³³⁸

The final challenge relates to laws that apply regardless of the provenance or nature of the data (such as those conditioned on the use of the data, the resultant outcome, or data quality), yet whose assumptions and internal balances do not necessarily fit the effects of synthetic data.

Let us separate the different cases. Some legal prohibitions, which are based on the assumption that a certain conduct is always welfare-reducing, do not require a different balance. Take, for example, consumer protection laws which prohibit data-based **deceptive practices**,³³⁹ or laws that prohibit certain types of data-based **bias**,³⁴⁰ whether directly or through fairness requirements.³⁴¹ Such laws apply based on the outcome, and thus capture both real and synthetic datasets. Prohibiting such conduct increases social welfare, regardless of the nature of the data.

Next, let us consider laws which apply regardless of the provenance of the data, yet synthetic data can change the overall effects on social welfare. We illustrate such challenges by laws that focus on **data quality** as a requirement for decision-making. For instance, some types of health data are subject to extensive quality standardization.³⁴² Data quality is also one of the core principles found in data privacy frameworks.³⁴³ For example, the Federal Privacy Act requires regulatory agencies to ensure that all records which are used in making any determination about an individual are made “with such accuracy... and completeness as is reasonably necessary to assure fairness to the individual in the determination”.³⁴⁴ Likewise, the accuracy of data is an

³³⁶ See, e.g., Gal & Rubinfeld, *supra* note 164, at 762-3.

³³⁷ Freeman et al., *supra* note 330, at 7.

³³⁸ Wang, Myles & Tucker, *supra* note 191, at 825 (Synthetic data must not be used to make clinical decisions. But, because the structure of the data is the same as the collected data, it can be used to plan and refine analyses before making a formal request to Public Health England’s Office for Data Release to conduct the same analysis on the collected data.).

³³⁹ Section 5(a) FTC Act declares “unfair or deceptive” acts unlawful. The FTC has suggested that it may use this provision in order to sanction deceptive data-based practices such as dark patterns. FED. TRADE COMM’N – STAFF REPORT, BRINGING DARK PATTERNS TO LIGHT, Sept. 2022, https://www.ftc.gov/system/files/ftc_gov/pdf/P214800%20Dark%20Patterns%20Report%209.14.2022%20-%20FINAL.pdf

³⁴⁰ See *infra* section IV.

³⁴¹ See, e.g., the requirement of fairness in Article 5(1)(a) GDPR which has been interpreted to prohibit bias. FUTURE OF PRIVACY FORUM (FPF), AUTOMATED DECISION-MAKING UNDER THE GDPR: PRACTICAL CASES FROM COURTS AND DATA PROTECTION AUTHORITIES 36, 39 (2022).

³⁴² Gal & Rubinfeld, *supra* note 164, at 740.

³⁴³ See, e.g., OECD, PRIVACY GUIDELINES, *supra* note 206, para. 8.

³⁴⁴ 5 U.S.C. § 552a(e)(5).

important component of the Fair Credit Reporting Act (FCRA). Any entity providing data about its customers to consumer reporting agencies (CRAs) for inclusion in a consumer report must provide accurate information.³⁴⁵ In addition, CRAs are under an obligation to “follow reasonable procedures to assure maximum possible accuracy of the information concerning the individual about whom the [consumer] report relates”.³⁴⁶

Such laws are based on the epistemic concern that where data is used as the basis for decision-making, the reliability of the decision will be affected by the quality of the underlying data.³⁴⁷ It is mostly assumed that improved data quality will increase social welfare. As Barocas observes, when speaking of decisions that affect individuals, “at issue is the simple fact that certain individuals may be subject to erroneous inferences” based on their data doubles.³⁴⁸ Of concern are harms and rights breaches, like denials of credit, social welfare rights, bail, or employment opportunities, as well as societal harms, such as entrenching existing misrepresentations and stereotypes, scaling miscarriages of justice, and exacerbating information and power asymmetries.³⁴⁹ As both state and private actors increasingly resort to data-informed decision-making across almost all areas of human activity, we might expect to see such accuracy requirements proliferate. Misrepresentation, underrepresentation (including by omission), or overrepresentation in a dataset are often viewed as examples of accuracy errors: what happens when things go wrong. As noted, synthetic data might improve compliance with such legal requirements.³⁵⁰

Yet, as noted above, not in all situations is increased quality welfare-enhancing. Indeed, the law does not promote data accuracy as an absolute value.³⁵¹ Some laws already recognize that, in some contexts, better data accuracy may have negative welfare effects. Consider, for instance, medical insurance. While an insurer might wish to have the most granular information possible about individuals seeking insurance in order to accurately assess the firm’s risk from insuring them, this more granular profiling will work to the detriment of some individuals (e.g., those who are predisposed to certain illnesses). The law often recognizes the merits of broad insurance coverage and limits the information that can be relied upon by insurers to calculate premiums.³⁵² In this sense, the law acts as a constraint on accuracy, to promote a better power balance between the relevant parties and to achieve broader social goals. Such laws apply to

³⁴⁵ 15 U.S.C. §1681.

³⁴⁶ 15 U.S. Code § 1681e(b).

³⁴⁷ Mittelstadt *et al*, *supra* note 2, at 5 (conclusions can only be as reliable and neutral as the data on which they are based).

³⁴⁸ Solon Barocas, *Data Mining and the Discourse on Discrimination*, GITHUB.IO (2014), <https://dataethics.github.io/proceedings/DataMiningandtheDiscourseOnDiscrimination.pdf>

³⁴⁹ Wachter and Mittelstadt, *supra* note 252.

³⁵⁰ Tordable, *supra* note 92 (A central goal of synthetic data is to overcome limitations and restrictions on obtaining real-world data by using artificially generated data that is similar to real-world data in a meaningful way).

³⁵¹ OECD, PRIVACY GUIDELINES, *supra* note 206, Clause 8. There is a difference between not requiring a firm to reach a high level of accuracy, and mandating it to artificially reduce the level of accuracy (e.g. by adding randomness to the dataset). The latter might also reduce accountability. Nielsen, *Bounding*, *supra* note 298.

³⁵² Louis DeNicola, *Which States Restrict the Use of Credit Scores in Determining Insurance Rates*, EXPERIAN (Sept. 23, 2020), <https://www.experian.com/blogs/ask-experian/which-states-prohibit-or-restrict-the-use-of-credit-based-insurance-scores/>.

both real and synthetic data.³⁵³ The level of data quality required by law may also affect such a balance. For example, completeness of a dataset is often required only to the extent necessary for the purposes of its processing.³⁵⁴ While such requirements may be based on cost–benefit efficiency considerations, they might also implicitly recognize that there is merit in obfuscation and incomplete datasets in some circumstances.³⁵⁵ Yet in most cases the law does not mandate data holders to limit or reduce the quality of their datasets.

In light of the above, supplementary data governance tools are required. A first step is to determine in which contexts the costs of higher accuracy for social welfare outweigh its benefits. This involves not only the identification of specific products and services, but also the level of accuracy at which the balance will tip in each case. The second stage focuses on determining which, if any, regulatory tools might best achieve such a balance, based on a comparative analysis of the costs and benefits of applying different tools, informed by the enforcement of accuracy-limiting tools that are already in place.³⁵⁶ Existing tools that can be used as potential sources of legal power include, inter alia, core principles found in data privacy law such as data security;³⁵⁷ antitrust prohibitions that regulate the ability to collect or generate data;³⁵⁸ or the use of the fair trade requirements included in the FTC Act to set bounds on the accuracy of predictive analytics.³⁵⁹ Yet we may need to adopt additional legal measures. Along these lines, Ohm suggests the creation of “throttling metrics,” by which friction in the algorithm might protect important human values,³⁶⁰ and Nielsen proposes that the accuracy of automated decision-making systems may be bounded where the output is too accurate for the context and leads to social harms.³⁶¹ The challenge lies in identifying those specific contexts in which such measures are justified, to ensure a welfare-increasing balance between accuracy and other societal goals.

Unfortunately, several recently proposed laws exemplify missed opportunities to acknowledge and take account of the need for such balancing.³⁶² The Algorithmic Accountability Act, proposed in 2019,³⁶³ which was designed to require assessments of

³⁵³ The findings of the UK Supreme Court in the case of *PJS* could be read in this way as providing some practical obscurity to the claimants by preventing further publication of private information even though it was already circulating in the public domain. *PJS v News Group Newspapers Ltd* [2016] UKSC 26.

³⁵⁴ OECD, PRIVACY GUIDELINES, *supra* note 206, Clause 8.

³⁵⁵ See also Genetic Information Nondiscrimination Act of 2008, 42 U.S.C. § 2000ff (prohibiting discrimination on the basis of genetic information in health insurance and employment scenarios, even if such data exists).

³⁵⁶ See examples throughout this section. For a more complete list see Nielsen, *supra* note 298.

³⁵⁷ OECD, PRIVACY GUIDELINES, *supra* note 206, Clause 11.

³⁵⁸ See, e.g., Dina Srinivasan, *The Antitrust Case Against Facebook: A Monopolist’s Journey Towards Pervasive Surveillance in Spite of Consumers’ Preference for Privacy* § II-C, 16 BERKELEY BUS. L.J. 39 (2019) (arguing that the monopoly rents that Facebook may have inflicted on consumers are in the form of pervasive surveillance).

³⁵⁹ Dennis D. Hirsch, *From Individual Control to Social Protection: New Paradigms for Privacy Law in the Age of Predictive Analytics*, 79 MD. L. REV. 439, 497 (2020).

³⁶⁰ Paul Ohm, *Throttling Machine Learning*, in LIFE AND LAW IN THE ERA OF DATA-DRIVEN AGENCY 214 (Mireille Hildebrandt & Kieron O’Hara eds., 2020). See also Talia B. Gillis & Jann L. Spiess, *Big Data and Discrimination*, 86 U. CHI. L. REV. 459 (2019).

³⁶¹ Nielsen, *supra* note 25.

³⁶² Nielsen, *Bounded*, *supra* note 298.

³⁶³ H.R. 2231, 116th Cong. (2019).

the costs and benefits of high-risk automated systems, focuses on the privacy and security of personal information.³⁶⁴ Likewise, the Algorithmic Justice and Online Platform Transparency Act of 2021³⁶⁵ incorporates an extremely narrow menu of tools to address algorithmic harms, which is limited to transparency, a right to data portability, and non-discrimination.³⁶⁶

VI. CONCLUSIONS

Synthetic data created a revolution in data generation. Its techniques have advanced to the point that in some instances it can replace collected datasets with fully or partially synthetic datasets characterized by a similar or even higher level of utility. Synthetic data has also brought about a qualitative shift, where fewer bits of collected data need to be combined to create learnings. While it is not a panacea, in some contexts it can significantly reduce access barriers to data, extend the scope of use of collected data, increase data quality, and reduce privacy and data security breaches. As such, it can be seen as a technological method for self-improvement of data-related decisions, and for overcoming some obstacles to the collection and use of data. It is thus not surprising that the use of synthetic data is becoming commonplace. Indeed, as noted above, most of the data used to train automated systems will soon be synthetic.

At the same time, synthetic data creates data governance challenges. This article focused on challenges resulting from three main effects of synthetic data: data access, data privacy, and data quality. As shown, synthetic data requires us to rethink the current legal status quo between data utility and data harms. Depending on the context, it could help alleviate or reinforce data-related social challenges, including fairness, equality, transparency, trust, and democracy, thereby illuminating many of the existing challenges in contemporary data governance.³⁶⁷ Some challenges are not new: synthetic data reflects or strengthens issues that are also likely to arise with regard to collected data. Yet synthetic data could significantly increase their prevalence. For example, by increasing data externalities and data-based collective harms, synthetic data strengthens the case for regulation which focuses on usage rather than on data provenance. Or it further blurs the lines between personal and non-personal data, calling into question the utility of this binary divide.³⁶⁸ Other challenges are unique to synthetic data. For instance, synthetic data challenges the ingrained assumption in some laws that firms need collected data to affect welfare. Yet in both cases the challenges created by synthetic data often go to the core of the legal data regime, mandating answers to questions like what are we protecting and why. Given its nature as a general-purpose technology, such effects are relevant across numerous industries.

³⁶⁴ Nielsen, *Bounded*, *supra* note 298.

³⁶⁵ S.1896, 117th Congress (2021-2).

³⁶⁶ *Id.*, Sections 3-5.

³⁶⁷ See, e.g., Ryan Calo, *Robotics and the Lessons of Cyberlaw*, 103 CAL. L. REV. 513, 534 (2015).

³⁶⁸ See, e.g., Nadezhda Purtova, *The law of Everything: Broad Concept of Personal Data and Future of EU Data Protection Law*, 10(1) L., INNOV. AND TECH. 40 (2018).

Some legal challenges identified are cross-sectional, and pertain to all three areas analyzed in the article, such as the need to reevaluate the level of risk to some rights (such as privacy, security, or non-discrimination) or motivating factors (such as dynamic efficiency), once synthetic data is used. As a result, some laws are mismatched with legal challenges. In others, a nuanced application might be required. One example pertains to the interpretation of reasonableness requirements. Synthetic data can raise the benchmark (e.g., where bias can be reduced by adding synthetic data), or lower it (such as a decision by a monopolist not to share collected data where synthetic data is comparable). Other challenges may pertain to one area affected by synthetic data.

As the synthetic data train has already left the station, it is surprising and even disconcerting that almost no attempts have been made in the legal literature to deal with such challenges, beyond the effects of synthetic data on privacy protection,³⁶⁹ or in the context of deep fakes.³⁷⁰ Furthermore, it seems that there is a disconnect between legal requirements and what firms in the industry, and academics writing in the business context, believe such requirements to be, and both sides currently disregard some of the most pertinent relevant legal challenges.³⁷¹ This article attempts to partially fill this gap. By doing so, it hopefully increases legal certainty for firms wishing to use synthetic data, and for decision-makers applying laws to synthetic data. It also hopefully paints a picture of requirements for legal change, to potentially increase social welfare.

We leave for future research challenges which arise when synthetic data is regulated in a dissimilar fashion in different states or jurisdictions. Indeed, synthetic data provides a good example of the stark chasm that exists between U.S. statutory data privacy law and the EU data privacy framework.³⁷² We also leave for future research some areas of law which were not covered in this article. These include, inter alia, whether certain property rights (such as copyrights)³⁷³ limit the use of collected data as a basis for creating synthetic data; to what extent does the First Amendment apply to inference data;³⁷⁴ how should deep fake images and fake profiles, based on synthetic data, be regulated;³⁷⁵ who owns computer creations (such as synthetic images); is it ethical to use one's medical profile to create a "virtual twin" to be used in virtual clinical trials;³⁷⁶ who

³⁶⁹ See *supra* Section IV.

³⁷⁰ Citron & Chesney, *supra* note 24, focusing on the negative effects. The use of deep fakes is not always negative. Katrina G. Geddes, *Ocularcentrism and Deepfakes: Should Seeing Be Believing?* 31(4) *FORDHAM IP, MEDIA & ENTERTAINMENT L. J.* 1042 (2021).

³⁷¹ See, e.g., *supra* Section IV.

³⁷² The EU and the U.S. seek to protect data privacy in different ways. See generally Schwartz & Solove, *supra* note 206; Chander et al, *supra* note 217.

³⁷³ Chloe Xiang, *AI Is Probably Using Your Images and It's Not Easy to Opt Out*, VICE (Sep. 26, 2022), <https://www.vice.com/en/article/3ad58k/ai-is-probably-using-your-images-and-its-not-easy-to-opt-out>. Some interesting issues include how fair use will be applied where images are used to train an AI system, and what should the remedy be when only a small part of the images used to train the algorithm are real, and their use is illegal.

³⁷⁴ Paul Ohm, *How to Regulate Harmful Inferences*, JOTWELL (Dec. 22, 2021), <https://cyber.jotwell.com/how-to-regulate-harmful-inferences/>.

³⁷⁵ See, e.g., Danielle Citron & Yitzchak Besser Chesney, *Web of Lies: Hate Speech, Pseudonyms, The Internet, Impersonator Trolls, and Fake Jews in the Era of Fake News*, 17 *OHIO STATE TECH. L. J.* 234 (2021); Solomon Asch, *Opinions and Social Pressure*, 193 *SCIENTIFIC AMERICAN* 31 (1955); Rex D. Glensy, *The Right to Dignity*, 43 *COLUM. HUMAN RIGHTS L. REV.* 65 (2011).

³⁷⁶ Proffitt, *supra* note 98.

is legally liable for harmful synthetic data; whether risk-based liability is better suited for synthetic data uses;³⁷⁷ and how synthetic data affects those laws that can otherwise help in reducing negative data externalities, such as contracts and disclosure law. While canvassing the effects of synthetic data on all areas of the law is beyond the scope of this article, these harms share many of the considerations elaborated above. Accordingly, we hope this article has provoked thought in these areas as well.

³⁷⁷ See generally Ira Rubinstein and Woodrow Hartzog, *Anonymization and Risk*, 91 WASH. L. REV. 703 (2016) (Suggesting a focus on the process of minimizing risk of reidentification and sensitive attribute disclosure, rather than on preventing harm).